

**AP EAMCET-2018-Engineering-22 Apr-Shift-1
Mathematics**

1. If $f: R \rightarrow R$ is defined by $f(x) = [2x] - 2[x]$ for $x \in R$, then the range of f is (Here $[x]$ denotes the greatest integer not exceeding x)

- (1) Z , the set of all integers
- (2) N , the set of all natural numbers
- (3) R , the set of all real numbers
- (4) $\{0, 1\}$

2. Given that a, b and c are real numbers such that $b^2 = 4ac$ and $a > 0$. The maximal possible set $D \subseteq R$ on which the function $f: D \rightarrow R$ given by,

$f(x) = \log \{ ax^3 + (a+b)x^2 + (b+c)x + c \}$ is defined, is

- (1) $R - \left\{ -\frac{b}{2a} \right\}$
- (2) $R - \left(\left\{ -\frac{b}{2a} \right\} \cup (-\infty, -1) \right)$

$$(3) R - \left(\left\{ -\frac{b}{2a} \right\} \cup \{x : x \geq 1\} \right)$$

$$(4) R - \left(\left\{ -\frac{b}{2a} \right\} \cup (-\infty, -1] \right)$$

3. For any natural number n , $(15 \times 5^{2n}) + (2 \times 2^{3n})$ is divisible by

- (1) 7
- (2) 11
- (3) 13
- (4) 17

4. For the matrix $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, $A^{-1} =$

- (1) A
- (2) A^2
- (3) A^3
- (4) A^4

5. If $A = \begin{bmatrix} k/2 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & m/4 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/4 \end{bmatrix}$, then

$$k + l + m =$$

- (1) 1
- (2) 9
- (3) 14
- (4) 29

6. If A and B are the two real values of k for which the system of equations $x + 2y + z = 1$, $x + 3y + 4z = k$, $x + 5y + 10z = k^2$ is consistent, then $A + B =$

- (1) 3
- (2) 4
- (3) 5
- (4) 7

7. Let $z = x + iy$ and point P represent z in the Argand plane. If the real part of $\frac{z-1}{z+i}$ is 1, then a point that lies on the locus of P is

- (1) (2016, 2017)
- (2) (-2016, 2017)
- (3) (-2016, -2017)
- (4) (2016, -2017)

8. If $13e^{i \tan^{-1} \frac{5}{12}} = a + ib$, then the ordered pair $(a, b) =$

- (1) (12, 5)
- (2) (5, 12)
- (3) (24, 10)
- (4) (10, 24)

9. If $z_1 = 1 - 2i$; $z_2 = 1 + i$ and $z_3 = 3 + 4i$, then $\left(\frac{1}{z_1} + \frac{3}{z_2}\right) \frac{z_3}{z_2} =$

- (1) $13 - 6i$
- (2) $13 - 3i$
- (3) $6 - \frac{13}{2}i$
- (4) $\frac{13}{2} - 3i$

10. If $1, \omega, \omega^2$ are the cube roots of unity, then

$$\frac{1}{1+2\omega} + \frac{1}{2+\omega} - \frac{1}{1+\omega} =$$

- (1) 1
- (2) ω
- (3) ω^2
- (4) 0

11. The number of integral values of x satisfying

$$5x - 1 < (x + 1)^2 < 7x - 3 \text{ is}$$

- (1) 0
- (2) 1
- (3) 2
- (4) 3

12. For real number x , if the minimum value of

$f(x) = x^2 + 2bx + 2c^2$ is greater than the maximum value of

$g(x) = -x^2 - 2cx + b^2$, then

- (1) $c^2 > 2b^2$
- (2) $c^2 < 2b^2$

$$(3) \ b^2 = 2c^2$$

$$(4) \ c^2 = 2b^2$$

13. If a , b and c are the roots of $x^3 + qx + r = 0$, then

$$(a-b)^2 + (b-c)^2 + (c-a)^2 =$$

$$(1) \ -6q$$

$$(2) \ -4q$$

$$(3) \ 6q$$

$$(4) \ 4q$$

14. If the sum of two roots of the equation $x^3 - 2px^2 + 3qx - 4r = 0$ is zero, then the value of r is

$$(1) \ \frac{3pq}{2}$$

$$(2) \ \frac{3pq}{4}$$

$$(3) \ pq$$

$$(4) \ 2pq$$

15. The sum of the four digit even numbers that can be formed with the digits 0, 3, 5, 4 without repetition is

- (1) 14684
- (2) 43536
- (3) 46526
- (4) 52336

16. If x is the number of ways in which six women and six men can be arranged to sit in a row such that no two women are together and if y is the number of ways they are seated around a table in the same manner, then $x:y =$

- (1) 12 : 1
- (2) 42 : 1
- (3) 16 : 1
- (4) 6 : 1

17. The number of 5-letter words that can be formed by using the letters of the word SARANAM is

- (1) 1120
- (2) 6720
- (3) 480

(4) 720

18. The number of rational terms in the binomial expansion of

$$\left(\sqrt[4]{5} + \sqrt[5]{4}\right)^{100} \text{ is}$$

(1) 50

(2) 5

(3) 6

(4) 51

19. The numerically greatest term in the binomial expansion of

$$(2a - 3b)^{19} \text{ when } a = \frac{1}{4} \text{ and } b = \frac{2}{3} \text{ is,}$$

(1) ${}^{19}C_5 \cdot 2^{11}$

(2) ${}^{19}C_3 \cdot \frac{1}{2^{11}}$

(3) ${}^{19}C_4 \cdot \frac{1}{2^{13}}$

(4) ${}^{19}C_3 \cdot 2^{13}$

20. If $\frac{x^2 + 5x + 7}{(x-3)^3} + \frac{A}{(x-3)} + \frac{B}{(x-3)^2} + \frac{C}{(x-3)^3}$, then the equation

of the line having slope A and passing through the point (B, C) is

(1) $x + y - 20 = 0$

(2) $x - y + 20 = 0$

(3) $x + y + 20 = 0$

(4) $x - y - 20 = 0$

21. If $\cos\left(x - \frac{\pi}{3}\right), \cos x, \cos\left(x + \frac{\pi}{3}\right)$ are in a harmonic progression,

then $\cos x =$

(1) $\frac{3}{2}$

(2) 1

(3) $\frac{\sqrt{3}}{2}$

(4) $\sqrt{\frac{3}{2}}$

22. $\cos^3 110^\circ + \cos^3 10^\circ + \cos^3 130^\circ =$

(1) $\frac{3}{4}$

(2) $\frac{3}{8}$

(3) $\frac{3\sqrt{3}}{8}$

(4) $\frac{3\sqrt{3}}{4}$

23. If the general solution of $\sin 5x = \cos 2x$ is of the form $a_n \cdot \frac{\pi}{2}$ for

$n = 0, \pm 1, \pm 2, \dots$, then $a_n =$

(1) $\frac{2n}{5 + 2(-1)^n}$

(2) $\frac{2n + (-1)^n}{5 + 2(-1)^n}$

(3) $\frac{2n + 1}{5 + 2(-1)^n}$

(4) $\frac{2n - 1}{5 + 2(-1)^n}$

24. Let x, y be real numbers such that $x \neq y$ and $xy \neq 1$. If

$$ax + b \sec(\tan^{-1} x) = c \text{ and } ay + b \sec(\tan^{-1} y) = c, \text{ then } \frac{x+y}{1-xy} =$$

(1) $\frac{2ab}{a^2 - b^2}$

(2) $\frac{2ac}{a^2 + c^2}$

(3) $\frac{2ab}{a^2 + b^2}$

(4) $\frac{2ac}{a^2 - c^2}$

25. $\tanh^{-1} \frac{1}{2} + \coth^{-1} 3 =$

(1) $\log \sqrt{6}$

(2) $\log 6$

(3) $-\log \sqrt{6}$

(4) $-\log 6$

26. If the median of a $\triangle ABC$ through A is perpendicular to AC , then

$$\frac{\tan A}{\tan C} =$$

(1) $1 + \sqrt{2}$

(2) $-\frac{1}{\sqrt{3}} + 1$

(3) -2

(4) $1 + \frac{2}{\sqrt{3}}$

27. In $\triangle ABC$, $\tan \frac{A}{2} + \tan \frac{B}{2} =$

(1) $\frac{c \cot \frac{C}{2}}{4s}$

(2) $\frac{2c \cot \frac{C}{2}}{a+b+c}$

(3) $\frac{2c \tan \frac{C}{2}}{s}$

(4) $\frac{c \cot \frac{C}{2}}{a+b+c}$

28. In a ΔABC , D , E and F respectively are the points of contact of the incircle with the sides AB , BC and CA such that $AD = \alpha$,

$BE = \beta$ and $CF = \gamma$, then $\frac{\alpha\beta\gamma}{\alpha + \beta + \gamma} =$

- (1) R^2
- (2) $2R$
- (3) $2r$
- (4) r^2

29. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-coplanar vectors. The vector equation of a line which passes through the point of intersection of two lines, one joining the points $\vec{a} + 2\vec{b} - 5\vec{c}$, $-\vec{a} - 2\vec{b} - 3\vec{c}$ and the other joining the points $-4\vec{c}$, $6\vec{a} - 4\vec{b} + 4\vec{c}$ is,

- (1) $\vec{r} = 2\vec{a} - 4\vec{b} + 3\vec{c} + \mu(\vec{a} - 6\vec{b} + 4\vec{c})$
- (2) $\vec{r} = 3\vec{a} + 6\vec{b} - \vec{c} + \mu(\vec{a} + 2\vec{b} + \vec{c})$
- (3) $\vec{r} = 2\vec{a} + 3\vec{b} - \vec{c} + \mu(\vec{a} + \vec{b} - \vec{c})$
- (4) $\vec{r} = -2\vec{b} + 3\vec{c} + \mu(\vec{a} - 4\vec{b} + 3\vec{c})$

30. In ΔPQR , M is the midpoint of QR and C is the midpoint of

PM . If QC when extended meets PR at N , then $\frac{|QN|}{|CN|} =$

- (1) 1
- (2) 2
- (3) 3
- (4) 4

31. If $\vec{a} = \hat{i} - 2\hat{j} - 3\hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{c} = \hat{i} + 3\hat{j} - 2\hat{k}$, then

$$\left[(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c}) (\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a}) (\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b}) \right] =$$

- (1) 160000
- (2) -8000
- (3) 400
- (4) -40

32. If $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j} - 2\hat{k}$, $\vec{n}$ is perpendicular to both $\vec{a}$ and $\vec{b}$ and θ is the angle between $\vec{c}$ and $\vec{n}$ then $\sin \theta =$

(1) $\sqrt{\frac{2}{3}}$

(2) $\frac{\sqrt{2}}{3\sqrt{3}}$

(3) $\frac{2}{\sqrt{3}}$

(4) $\frac{\sqrt{3}}{2}$

33. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular vectors of the same magnitude, then the cosine of the angle between $\vec{a}$ and $\vec{a} + \vec{b} + \vec{c}$ is

(1) $\frac{1}{\sqrt{2}}$

(2) $\frac{1}{\sqrt{3}}$

(3) $\frac{1}{2}$

(4) $\frac{\sqrt{3}}{2}$

34. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-coplanar vectors and the four points with position vectors $2\vec{a} + 3\vec{b} - \vec{c}$, $\vec{a} - 2\vec{b} + 3\vec{c}$, $3\vec{a} + 4\vec{b} + 2\vec{c}$ and $k\vec{a} - 6\vec{b} + 6\vec{c}$ are coplanar, then $k =$

- (1) 0
- (2) 1
- (3) 2
- (4) 3

35. The mean and the standard deviation of a data of 8 items are 25 and 5 respectively. If two items 15 and 25 are added to this data, then the variance of the new data is

- (1) 29
- (2) 24
- (3) 26
- (4) $\sqrt{29}$

36. The mean deviation from the median for the following distribution f collected to two decimals) is

x_i	6	9	3	12	15	13	21	22
f_i	4	5	3	2	5	4	4	3

- (1) 13.42
- (2) 5.45
- (3) 4.97
- (4) 11.25

37. If a die is rolled three times, then the probability of getting a larger number on its face than the previous number each time, is

- (1) $\frac{15}{216}$
- (2) $\frac{5}{54}$
- (3) $\frac{13}{216}$
- (4) $\frac{1}{18}$

38. A man is known to speak the truth 2 out of 3 times, If he throws a die and reports that it is six, then the probability that it is actually five, is

- (1) $\frac{3}{8}$

(2) $\frac{1}{7}$

(3) $\frac{2}{7}$

(4) $\frac{4}{5}$

39. If the probability function of a random variable X is defined by

$$P(X = k) = a \left(\frac{k+1}{2^k} \right) \text{ for } k = 0, 1, 2, 3, 4, 5 \text{ then the probability}$$

that X takes a prime value is

(1) $\frac{13}{20}$

(2) $\frac{23}{60}$

(3) $\frac{11}{20}$

(4) $\frac{19}{60}$

40. If X is a binomial variate with mean 6 and variance 2, then the value of $P(5 \leq X \leq 7)$ is

(1) $\frac{4762}{6561}$

(2) $\frac{4672}{6561}$

(3) $\frac{5264}{6561}$

(4) $\frac{5462}{6651}$

41. Let $A(2, 3)$, $B(3, -6)$, $C(5, -7)$ be three points. If P is a point satisfying the condition $PA^2 + PB^2 = 2PC^2$, then a point that lies on the locus of P is

(1) $(2, -5)$

(2) $(-2, 5)$

(3) $(13, 10)$

(4) $(-13, -10)$

42. If the coordinates of a point P changes to $(2, -6)$ when the coordinate axes are rotated through an angle of 135° , then the coordinates of P in the original system are

- (1) $(-2, 6)$
- (2) $(-6, 2)$
- (3) $(2\sqrt{2}, 4\sqrt{2})$
- (4) $(\sqrt{2}, -\sqrt{2})$

43. If the portion of a line intercepted between the coordinate axes is divided by the point $(2, -1)$ in the ratio 3:2, then the equation of that line is

- (1) $5x - 2y - 20 = 0$
- (2) $2x - y - 5 = 0$
- (3) $3x - y - 7 = 0$
- (4) $x - 3y - 5 = 0$

44. The equation of the line passing through the point of intersection of the lines $2x + y - 4 = 0$; $x - 3y + 5 = 0$ and lying at a distance of $\sqrt{5}$ units from the origin, is

- (1) $x - 2y - 5 = 0$
- (2) $x + 2y - 5 = 0$

(3) $x + 2y + 5 = 0$

(4) $x - 2y + 5 = 0$

45. The equation of the line joining the centroid with the orthocentre of the triangle formed by the points $(-2, 3), (2, -1), (4, 0)$ is

(1) $x + y - 2 = 0$

(2) $11x - y - 14 = 0$

(3) $x - 11y + 6 = 0$

(4) $2x - y - 2 = 0$

46. The lines are represented by the equations $23x^2 - 48xy + 3y^2 = 0$ and $2x + 3y + 4 = 0$ form

(1) an isosceles triangle

(2) a right angled triangle

(3) an equilateral triangle

(4) a scalene triangle

47. If the line $x + 2y = k$ intersects the curve $x^2 - xy + y^2 + 3x + 3y - 2 = 0$ at two points A and B and if O is the origin, then the condition for $\angle AOB = 90^\circ$ is

(1) $k^2 + k + 1 = 0$

(2) $k^2 - 2k + 10 = 0$

(3) $2k^2 + 9k - 10 = 0$

(4) $3k^2 + 8k - 1 = 0$

48. If $2x^2 + 3xy - 2y^2 = 0$ represent two sides of a parallelogram and $3x + y + 1 = 0$ is one of its diagonals, then the other diagonal is

(1) $x - 3y + 1 = 0$

(2) $x - 3y + 2 = 0$

(3) $x - 3y = 0$

(4) $3x - y = 0$

49. If the lengths of the tangents drawn from P to the circles $x^2 + y^2 - 2x + 4y - 20 = 0$ and $x^2 + y^2 - 2x - 8y + 1 = 0$ are in the ratio $2 : 1$, then the locus P is

(1) $x^2 + y^2 + 2x + 12y + 8 = 0$

(2) $x^2 + y^2 - 2x + 12y + 8 = 0$

(3) $x^2 + y^2 + 2x - 12y + 8 = 0$

(4) $x^2 + y^2 - 2x - 12y + 8 = 0$

50. The equation of a circle touching the coordinate axes and the line $3x - 4y = 12$ is

(1) $x^2 + y^2 + 6x + 6y + 9 = 0$

(2) $x^2 + y^2 + 6x + 6y - 9 = 0$

(3) $x^2 + y^2 - 6x - 6y + 9 = 0$

(4) $x^2 + y^2 - 6x - 6y - 9 = 0$

51. The pole of the straight line $9x + y - 28 = 0$ with respect to the circle $2x^2 + 2y^2 - 3x + 5y - 7 = 0$ is

(1) $(3, 1)$

(2) $(3, -1)$

(3) $(-3, 1)$

(4) $(4, -8)$

52. The point of intersection of the direct common tangents drawn to the circles $(x + 11)^2 + (y - 2)^2 = 225$ and $(x - 11)^2 + (y + 2)^2 = 25$ is

(1) $\left(-\frac{11}{2}, 1\right)$

(2) $(-22, 4)$

(3) $\left(\frac{11}{2}, -1\right)$

(4) $(22, -4)$

53. In List-I, a pair of circles is given in A, B, C and in List-II, angle between those pair of circles is given. Match the items from List-I to List-II.

List-I

List-II

(A) $(x-2)^2 + y^2 = 2(x-2)^2 + (y-1)^2 = 1$

(I) 90°

(B) $x^2 + y^2 - 6x - 6y + 9 = 0$

(II) 135°

$x^2 + y^2 - 4x + 4y - 9 = 0$

(C) $x^2 + y^2 + 4x - 14y + 28 = 0$

(III) 60°

$x^2 + y^2 + 4x - 5 = 0$

(IV) 30°

The correct matching is

(1) A B C
I II III

(2) A B C
II I III

(3)

A	B	C
III	I	IV

(4)

A	B	C
IV	III	I

54. If the radical axis of the circles $x^2 + y^2 + 2gx + 2fy + c = 0$ and $2x^2 + 2y^2 + 3x + 8y + 2c = 0$ touches the circle $x^2 + y^2 + 2x + 2y + 1 = 0$, then

(1) $g = \frac{3}{4}$ or $f = 2$

(2) $g \neq \frac{3}{4}$ or $f = 2$

(3) $g = \frac{3}{4}$ or $f \neq 2$

(4) $g = \frac{2}{5}$ or $f = 1$

55. The line $y = 6x + 1$ touches the parabola $y^2 = 24x$. The coordinates of a point P on this line, from which the tangent to $y^2 = 24x$ is perpendicular to the line $y = 6x + 1$, is

(1) $(-1, -5)$

(2) $(-2, -11)$

(3) $(-6, -35)$

(4) $(-7, -41)$

56. A point on the parabola whose focus is $S(1, -1)$ and whose vertex is $A(1, 1)$ is

(1) $\left(3, \frac{1}{2}\right)$

(2) $(1, 2)$

(3) $\left(2, \frac{1}{2}\right)$

(4) $(2, 2)$

57. An ellipse having the coordinate axes as its axes and its major axis along Y -axis, passes through the point $(-3, 1)$ and has

eccentricity $\sqrt{\frac{2}{5}}$. Then its equation is

(1) $3x^2 + 5y^2 - 15 = 0$

(2) $5x^2 + 3y^2 - 32 = 0$

$$(3) 3x^2 + 5y^2 - 32 = 0$$

$$(4) 5x^2 + 3y^2 - 48 = 0$$

58. The product of the perpendicular distances drawn from the points $(3, 0)$ and $(-3, 0)$ to the tangent of the ellipse $\frac{x^2}{36} + \frac{y^2}{27} = 1$

at $\left(3, \frac{9}{2}\right)$ is

(1) 36

(2) 27

(3) 9

(4) 63

59. The equation of the hyperbola whose asymptotes are the lines $3x + 4y - 2 = 0$, $2x + y + 1 = 0$ and which passes through the point $(1, 1)$ is

(1) $6x^2 + 11xy + 4y^2 - 30x + 2y + 7 = 0$

(2) $6x^2 + 11xy + 4y^2 - x + 2y - 22 = 0$

(3) $6x^2 + 11xy + 4y^2 - x + 2y + 22 = 0$

(4) $6x^2 + 11xy + 4y^2 - 3x - 7y - 11 = 0$

60. If the orthocentre and the centroid of a triangle are $(-3, 5, 2)$ and $(3, 3, 4)$ respectively, then its circumcentre is
- (1) $(6, 2, 5)$
 - (2) $(6, 2, -5)$
 - (3) $(6, -2, 5)$
 - (4) $(6, -2, -5)$
61. A plane cuts the coordinate axes X, Y, Z at A, B, C respectively such that the centroid of the $\triangle ABC$ is $(6, 6, 3)$. Then the equation of that plane is
- (1) $x + y + z - 6 = 0$
 - (2) $x + 2y + z - 18 = 0$
 - (3) $2x + y + z - 18 = 0$
 - (4) $x + y + 2z - 18 = 0$
62. If the foot of the perpendicular drawn from the origin to a plane is $(1, 2, 3)$, then a point on that plane is

(1) $(3, 2, 1)$

(2) $(7, 2, 1)$

(3) $(7, 3, -1)$

(4) $(6, -3, 4)$

63. If $[x]$ denotes the greatest integer $\leq x$, then

$$\lim_{n \rightarrow \infty} \frac{1}{n^3} \{ [1^2 x] + [2^2 x] + [3^2 x] + \dots + [n^2 x] \} =$$

(1) $\frac{x}{2}$

(2) $\frac{x}{3}$

(3) $\frac{x}{6}$

(4) 0

64. If a function f is defined by

$$f(x) = \begin{cases} \frac{1 - \sqrt{2} \sin x}{\pi - 4x}, & \text{if } x \neq \frac{\pi}{4} \\ k, & \text{if } x = \frac{\pi}{4} \end{cases} \text{ is continuous at } x = \frac{\pi}{4}, \text{ then } k =$$

(1) $\frac{1}{4}$

(2) 1

(3) $-\frac{1}{4}$

(4) 2

65. The derivative of $f(x) = x^{\tan^{-1} x}$ with respect to

$g(x) = \sec^{-1}\left(\frac{1}{2x^2 - 1}\right)$ is

(1) $\frac{1}{2}\sqrt{1-x^2}x^{\tan^{-1} x}\left[\frac{\log x}{1+x^2} + \frac{\tan^{-1} x}{x}\right]$

(2) $\frac{1}{2}\sqrt{1-x^2}x^{\tan^{-1} x}\left[\log(\tan^{-1} x) + x(1+x^2)\tan^{-1} x\right]$

(3) $\frac{-2\tan^{-1} x\left[\frac{\log x}{1+x^2} + \frac{\tan^{-1} x}{x}\right]}{\sqrt{1-x^2}}$

(4) $-\frac{1}{2}\sqrt{1-x^2}x^{\tan^{-1} x}\left[\frac{\log x}{1+x^2} + \frac{\tan^{-1} x}{x}\right]$

66. If $x = 3 \cos t$ and $y = 4 \sin t$, then $\frac{d^2y}{dx^2}$ at the point

$(x_0, y_0) = \left(\frac{3}{2}\sqrt{2}, 2\sqrt{2}\right)$, is

(1) $\frac{4\sqrt{2}}{9}$

(2) $-\frac{4\sqrt{2}}{9}$

(3) $\frac{8\sqrt{2}}{9}$

(4) $-\frac{8\sqrt{2}}{9}$

67. If $y = \frac{2}{\sqrt{a^2 - b^2}} \tan^{-1} \left[\sqrt{\frac{a-b}{a+b}} \tan \frac{x}{2} \right]$, then $\frac{d^2y}{dx^2} \Big|_{x=\frac{\pi}{2}} =$

(1) $\frac{b}{2a^2}$

(2) $\frac{b}{a^2}$

(3) $\frac{2b}{a}$

(4) $\frac{b^2}{2a}$

68. If $f(x) = x^3 + ax^2 + bx + 5\sin^2 x$ is an increasing function on R , then

(1) $a^2 - 3b - 15 < 0$

(2) $a^2 - 3b + 15 > 0$

(3) $a^2 - 3b - 15 > 0$

(4) $a^2 + 3b + 15 > 0$

69. The approximate value of $\cos 31^\circ$ is

(Take $1^\circ = 0.0174$)

(1) 0.7521

(2) 0.866

(3) 0.7146

(4) 0.8573

70. If x and y are two positive numbers such that $x + y = 32$, then the minimum value of $x^2 + y^2$ is,

(1) 500

- (2) 256
- (3) 1024
- (4) 512

71. The constant 'c' of Lagrange's mean value theorem for the function $f(x) = \frac{2x+3}{4x-1}$ defined on $[1, 2]$ is

- (1) $\frac{1+\sqrt{15}}{3}$
- (2) $\frac{1+\sqrt{21}}{4}$
- (3) $\frac{5}{3}$
- (4) $\frac{3}{2}$

72. $\int \frac{\sin 2x dx}{\sin^4 x + \cos^4 x} = \tan^{-1}(f(x)) + c$, then $f\left(\frac{\pi}{3}\right) =$

- (1) 1
- (2) 2
- (3) 3

(4) $\frac{1}{3}$

73. $\int \left(\frac{\log x - 1}{1 + (\log x)^2} \right)^2 dx =$

(1) $\frac{\log x}{1 + (\log x)^2} + c$

(2) $\frac{x}{x^2 + 1} + c$

(3) $\frac{x}{1 + (\log x)^2} + c$

(4) $\frac{-x}{1 + (\log x)^2} + c$

74. $\int \frac{dx}{x^3 + 3x^2 + 2x} =$

(1) $\log|x| + \log\left|\frac{x+2}{x+1}\right| + c$

(2) $\log|x| - \log|x+1| + \log|x+2| + c$

(3) $\frac{1}{2} [\log|x| + \log|x+1| + \log|x+2|] + c$

$$(4) \frac{1}{2} \log \left(\frac{|x^2 + 2x|}{(x+1)^2} \right) + c$$

75. For $n \geq 2$, If $I_n = \int \sec^n x dx$, then $I_4 - \frac{2}{3} I_2 =$

(1) $\sec^2 x \tan x + c$

(2) $\frac{1}{3} \sec^2 x \tan x + c$

(3) $\frac{2}{3} \sec^2 x \tan x + c$

(4) $\frac{1}{2} \log |\sec x + \tan x| + c$

76. $\lim_{n \rightarrow \infty} \left(\frac{\sqrt{1} + 2\sqrt{2} + 3\sqrt{3} + \dots + n\sqrt{n}}{n^{5/2}} \right) =$

(1) 1

(2) $\frac{5}{2}$

(3) 0

(4) $\frac{2}{5}$

77. $\int_0^{\alpha/3} \frac{f(x)}{f(x) + f\left(\frac{\alpha - 3x}{3}\right)} dx =$

(1) $\frac{2\alpha}{3}$

(2) $\frac{\alpha}{2}$

(3) $\frac{\alpha}{3}$

(4) $\frac{\alpha}{6}$

78. The area (in sq. units) of the region bounded by the X -axis and the curve $y = 1 - x - 6x^2$ is

(1) $\frac{125}{216}$

(2) $\frac{125}{512}$

(3) $\frac{25}{216}$

(4) $\frac{25}{512}$

79. If m and n are respectively the order and degree of the differential equation of the family of parabolas with focus at the origin and X -axis as its axis, then $mn - m + n =$

- (1) 1
- (2) 4
- (3) 3
- (4) 2

80. The general solution of $\left(1 + e^{\frac{x}{y}}\right)dx + e^{\frac{x}{y}}\left(1 - \frac{x}{y}\right)dy = 0$ is

- (1) $ye^{y/x} + x = c$
- (2) $ye^{x/y} - x = c$
- (3) $ye^{x/y} + y = c$
- (4) $ye^{x/y} + x = c$

**AP EAMCET-2018-Engineering-22 Apr-Shift-1
Physics**

81. Two resistances $60.36 \, \Omega$ and $30.09 \, \Omega$ are connected in parallel.

The equivalent resistance is

(1) $20 \pm 0.08 \, \Omega$

(2) $20 \pm 0.06 \, \Omega$

(3) $20 \pm 0.03 \Omega$

(4) $20 \pm 0.10 \Omega$

82. Assertion (A): The velocity of a projectile at a point on its trajectory is equal to the slope at that point

Reason (R): The velocity vector at a point is always along the tangent to the trajectory at that point.

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true and R is not the correct explanation of A.
- (3) A is true but R is false
- (4) A is false but R is true.

83. A body is projected from the ground at an angle of $\tan^{-1}\left(\frac{8}{7}\right)$ with the horizontal. The ratio of the maximum height attained by it to its range is

- (1) 8:7
- (2) 4:7
- (3) 2:7
- (4) 1:7

84. A body is projected with a speed 'u' at an angle ' θ ' with the horizontal. The radius of curvature of the trajectory when it makes an angle $\left(\frac{\theta}{2}\right)$ with the horizontal is

(g - acceleration due to gravity)

(1) $\frac{u^2 \cos^2 \theta \sec^3 \left(\frac{\theta}{2}\right)}{\sqrt{3}g}$

(2) $\frac{u^2 \cos^2 \theta \sec^3 \left(\frac{\theta}{2}\right)}{2g}$

(3) $\frac{2u^2 \cos^3 \theta \sec^2 \left(\frac{\theta}{2}\right)}{g}$

(4) $\frac{u^2 \cos^2 \theta \sec^3 \left(\frac{\theta}{2}\right)}{g}$

85. Sand is to be piled up on a horizontal ground in the form of a regular cone of a fixed base of radius R . Coefficient of static friction between the sand layers is μ . Maximum volume of the

sand that can be piled up in the form of cone without slipping on the ground is

(1) $\frac{\mu R^3}{3\pi}$

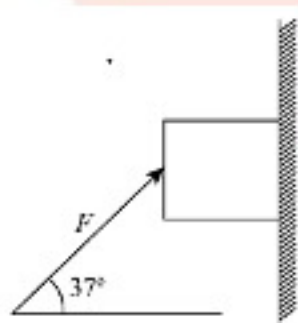
(2) $\frac{\mu R^3}{3}$

(3) $\frac{\pi R^3}{3\mu}$

(4) $\frac{\mu \pi R^3}{3}$

86. A block of mass 2 kg is being pushed against a wall by a force $F = 90$ N as shown in the figure. If the coefficient of friction is 0.25, then the magnitude of acceleration of the block is

(Take, $g = 10 \text{ ms}^{-2}$) $\left(\sin 37^\circ = \frac{3}{5} \right)$



- (1) 16 ms^{-2}
- (2) 8 ms^{-2}
- (3) 38 ms^{-2}
- (4) 54 ms^{-2}

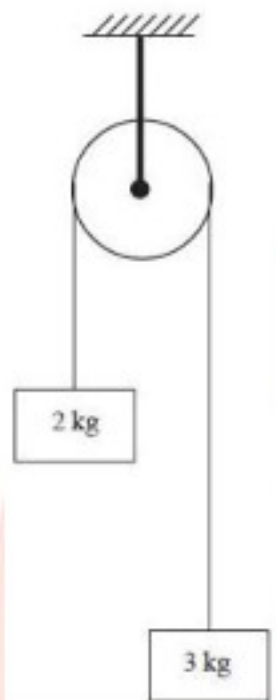
87. A body of mass 2 kg thrown vertically upward from the ground with a velocity of 8 ms^{-1} reaches a maximum height of 3 m. The work done by the air resistance is _____

(Acceleration due to gravity = 10 ms^{-2})

- (1) 4J
- (2) 60J
- (3) 64J
- (4) 8J

88. The system, of two masses 2 kg and 3 kg shown in the figure is released from rest. The work done on 3 kg block by the force of gravity during first 2 seconds of its motion is

($g = 10 \text{ ms}^{-2}$)



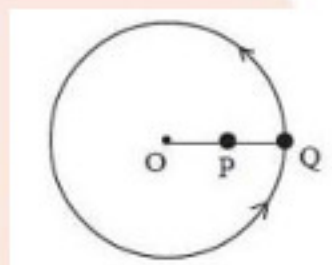
- (1) 120J
- (2) 80J
- (3) 40J
- (4) 30J

89. A rigid metallic sphere is spinning around its own axis in the absence of external torque. If the temperature is raised, its volume increases by 9%. The change in its angular speed is

- (1) increases by 9%
- (2) decreases by 9%

- (3) increases by 6%
- (4) decreases by 6%

90. Two spheres P and Q, each of mass 200 g are attached to a string of length one metre as shown in the figure. The string and the spheres are then whirled in a horizontal circle about O at a constant angular speed. The ratio of the tension in the string between P and Q to that of between P and O is (P is at the mid point of line joining O and Q)



- (1) $\frac{1}{2}$
- (2) $\frac{2}{3}$
- (3) $\frac{3}{2}$
- (4) $\frac{2}{1}$

91. The potential energy of a simple harmonic oscillator of mass 2 kg at its mean position is 5 J. If its total energy is 9 J and amplitude is 1 cm, then its time period is

- (1) $\frac{\pi}{100}$ s
- (2) $\frac{\pi}{50}$ s
- (3) $\frac{\pi}{20}$ s
- (4) $\frac{\pi}{10}$ s

92. Three masses m , $2m$ and $3m$ are arranged in two triangular configurations as shown in figure 1 and figure 2. Work done by an external agent in changing, the configuration from figure 1 to figure 2 is

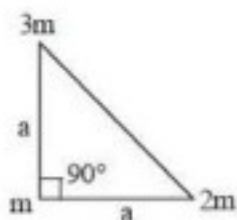


figure 1

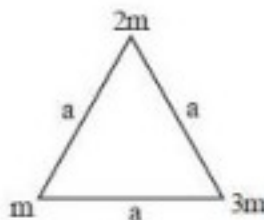


figure 2

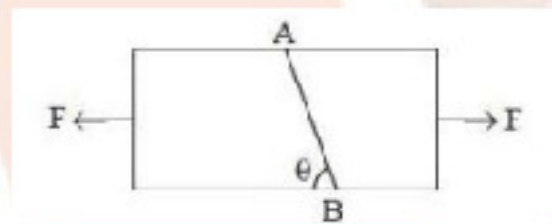
$$(1) \frac{6Gm^2}{a} \left[2 - \frac{6}{\sqrt{2}} \right]$$

$$(2) 0$$

$$(3) -\frac{Gm^2}{a} \left[6 + \frac{6}{\sqrt{2}} \right]$$

$$(4) -\frac{Gm^2}{a} \left[6 - \frac{6}{\sqrt{2}} \right]$$

93. Two equal and opposite forces each F act on a rod of uniform cross-sectional area ' a ' as shown in the figure. Shearing stress on the section AB will be



$$(1) \frac{F \sin \theta \cos \theta}{a}$$

$$(2) \frac{F \sin \theta}{a}$$

$$(3) \frac{F \cos \theta}{a}$$

$$(4) \frac{F \sin^2 \theta}{a}$$

94. A body is suspended by a light string. The tensions in the string when the body is in air, when the body is totally immersed in water and when the body is totally immersed in a liquid are respectively 40.2 N, 28.4 N and 16.6 N. The density of the liquid is

- (1) 1200 kg-m^{-3}
- (2) 1600 kg-m^{-3}
- (3) 2000 kg-m^{-3}
- (4) 2400 kg-m^{-3}

95. Steam at 100°C is passed into 1 kg of water contained in a calorimeter at 9°C till the temperature of water and calorimeter is increased to 90°C . The mass of the steam condensed is nearly

(Water equivalent of calorimeter = 0.1 kg, Specific heat of water = $1 \text{ cal g}^{-1}^\circ\text{C}^{-1}$ and latent heat of vapourisation = 540 cal g^{-1})

- (1) 81 g

- (2) 162 g
- (3) 243 g
- (4) 486 g

96. Three very large plates of same area are kept parallel and close to each other. They are considered as ideal black surfaces and have very high thermal conductivity. First and third plates are maintained at absolute temperatures $2T$ and $3T$ respectively. Temperature of the middle plate in steady state is

- (1) $\left(\frac{65}{2}\right)^{\frac{1}{4}} T$
- (2) $\left(\frac{97}{4}\right)^{\frac{1}{4}} T$
- (3) $\left(\frac{97}{2}\right)^{\frac{1}{4}} T$
- (4) $(97)^{\frac{1}{4}} T$

97. A thermally insulated vessel with nitrogen gas at 27°C is moving with a velocity of 100 ms^{-1} . If the vessel is stopped

suddenly, the percentage change in the pressure of the gas is nearly

(Assume entire loss in KE of the gas is given as heat to gas and $R = 8.3 \text{ Jmol}^{-1}\text{K}^{-1}$)

- (1) 1.1
- (2) 0.93
- (3) 0.5
- (4) 2.25

98. Match the following.

List-I

- A) Zeroth law of thermodynamics
- B) First law of thermodynamics
- C) Free expansion of gas
- D) Second law of thermodynamics

List-II

- I) Direction of flow of heat
- II) Work done is zero
- III) Thermal equilibrium
- IV) Law of conservation of energy

The correct answer is

	A	B	C	D
(1)	II	IV	III	I

	A	B	C	D
(2)	III	IV	II	I

	A	B	C	D
(3)	III	I	II	IV

	A	B	C	D
(4)	I	III	IV	II

99. For a molecule of an ideal gas, the number density is $2\sqrt{2} \times 10^8 \text{ cm}^{-3}$ and the mean free path is $\frac{10^{-2}}{\pi} \text{ cm}$. The diameter of the gas molecule is

- (1) $5 \times 10^{-4} \text{ cm}$
- (2) $0.5 \times 10^{-4} \text{ cm}$
- (3) $2.5 \times 10^{-4} \text{ cm}$
- (4) $4 \times 10^{-4} \text{ cm}$

100. A solid ball is suspended from the ceiling of a motor car through a light string. A transverse pulse travels at the speed 60

cm^{-1} on the string, when the car is at rest. When the car accelerates on a horizontal road, speed of the pulse is 66 cm^{-1} . The acceleration of the car is nearly ($g=10 \text{ ms}^{-2}$)

- (1) 4.3 ms^{-2}
- (2) 2.9 ms^{-2}
- (3) 6.8 ms^{-2}
- (4) 5.5 ms^{-2}

101. A reflector is moving with 20 ms^{-1} towards a stationary source of sound. If the source is producing sound waves of 160 Hz then the wavelength of the reflected wave is (speed of sound in air is 340 ms^{-1})

- (1) $\frac{17}{8} \text{ m}$
- (2) $\frac{17}{11} \text{ m}$
- (3) $\frac{17}{9} \text{ m}$
- (4) $\frac{17}{16} \text{ m}$

102. A light ray incidents normally on one surface of an equilateral prism. The angle of deviation of the light ray is

(Refractive index of the material of the prism = $\sqrt{2}$)

- (1) 60°
- (2) 30°
- (3) 0°
- (4) 120°

103. Two polaroids are placed in the path of unpolarised light beam of intensity I_0 such that no light is emitted from the second polaroid. If a third polaroid whose polarization axis makes an angle θ with that of the first polaroid is placed between the polaroids, then intensity of light emerging from the last polaroid is

- (1) $\left(\frac{I_0}{8}\right)\sin^2 2\theta$
- (2) $\left(\frac{I_0}{4}\right)\sin^2 2\theta$
- (3) $\left(\frac{I_0}{2}\right)\cos^2 \theta$

(4) $I_0 \cos^2 \theta$

104. Two point charges are kept in air with a separation between them. The force between them is F_1 , if half of the space between the charges is filled with a dielectric of dielectric constant 4 and the force between them is F_2 . If $\frac{1}{3}$ rd of the space between the charges is filled with dielectric of dielectric constant 9. Then $\frac{F_1}{F_2}$ is

- (1) $\frac{27}{64}$
- (2) $\frac{16}{81}$
- (3) $\frac{81}{64}$
- (4) $\frac{100}{81}$

105. A simple pendulum with a bob of mass 40 g and charge $+2 \mu\text{C}$ makes 20 oscillations in 44 s. A vertical electric field magnitude

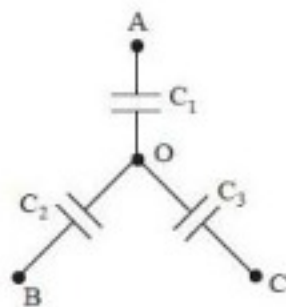
$4.2 \times 10^4 \text{ NC}^{-1}$ pointing downward is applied. The time taken by the pendulum to make 15 oscillations in the electric field is (acceleration due to gravity = 10 ms^{-2})

- (1) 30 s
- (2) 60 s
- (3) 90 s
- (4) 15 s

106. A parallel plate capacitor has a capacity $80 \times 10^{-6} \text{ F}$ when air is present between its plates. The space between the plates is filled with a dielectric slab of dielectric constant 20. The capacitor is now connected to a battery of 30 V by wires. The dielectric slab is then removed. Then the charge passing through the wire is

- (1) $12 \times 10^{-3} \text{ C}$
- (2) $25.3 \times 10^{-3} \text{ C}$
- (3) $120 \times 10^{-3} \text{ C}$
- (4) $45.6 \times 10^{-3} \text{ C}$

107. Three uncharged capacitors of capacities C_1 , C_2 and C_3 are connected as shown in the figure. A, B and C are at potentials V_1 , V_2 and V_3 , respectively. The potential at O is



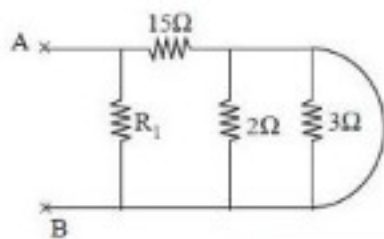
(1) $\frac{C_1 V_1 + C_2 V_2 + C_3 V_3}{C_1 + C_2 + C_3}$

(2) $\frac{C_1 V_1 + C_2 V_2 - C_3 V_3}{C_1 + C_2 + C_3}$

(3) $\frac{C_1 V_1 - C_2 V_2 - C_3 V_3}{C_1 + C_2 + C_3}$

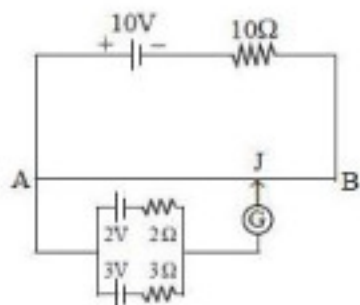
(4) zero

108. The equivalent resistance between A and B is 6 Ω . The value of R_1 is



- (1) $20\ \Omega$
- (2) $10\ \Omega$
- (3) $5\ \Omega$
- (4) $25\ \Omega$

109. A battery of emf 10 V is connected to a uniform wire AB of 1 m length and having a resistance of $10\ \Omega$ in series with a $10\ \Omega$ resistor as shown in the figure. Two cells of emf 2 V and 3 V having internal resistances $2\ \Omega$ and $3\ \Omega$ respectively are connected as shown in the figure. If the galvanometer shows null deflection at point J on the wire, the distance of point J from the point B is _____



- (1) 48 cm
- (2) 50 cm
- (3) 52 cm
- (4) 54 cm

110. Two infinitely long wires carry currents 4 A and 3 A placed along X -axis and Y -axis respectively. Magnetic field at a point $P(0,0,d)$ m will be _____ T.

- (1) $\frac{4\mu_0}{2\pi d}$
- (2) $\frac{3\mu_0}{2\pi d}$
- (3) $\frac{7\mu_0}{2\pi d}$
- (4) $\frac{5\mu_0}{2\pi d}$

111. Two moving coil galvanometers. X and Y have coils with resistances $10\ \Omega$ and $14\ \Omega$, cross sectional areas $4.8 \times 10^{-3}\ \text{m}^2$ and $2.4 \times 10^{-3}\ \text{m}^2$, number of turns 30 and 45 respectively. They are placed in magnetic fields of $0.25\ \text{T}$ and $0.50\ \text{T}$ respectively. Then the ratio of their current sensitivities and the ratio of their voltage sensitivities are respectively.

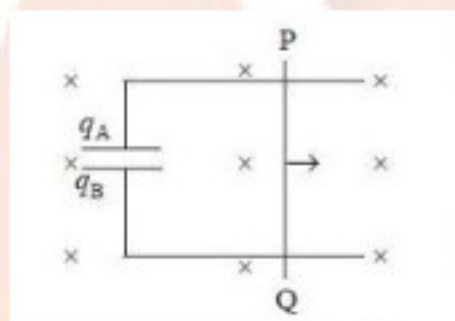
- (1) 2:3; 14:15
- (2) 5:7; 2:1
- (3) 2:13; 1:2
- (4) 14:15; 2:9

112. Two short bar magnets each of magnetic moment $9\ \text{Am}^2$ are placed such that one is at $x = -3\ \text{cm}$ and the other at $y = -3\ \text{cm}$. If their magnetic moments are directed along positive and negative X -directions respectively then the resultant magnetic field at the origin is

- (1) $100\ \text{T}$
- (2) $10\ \text{T}$
- (3) $0.1\ \text{T}$

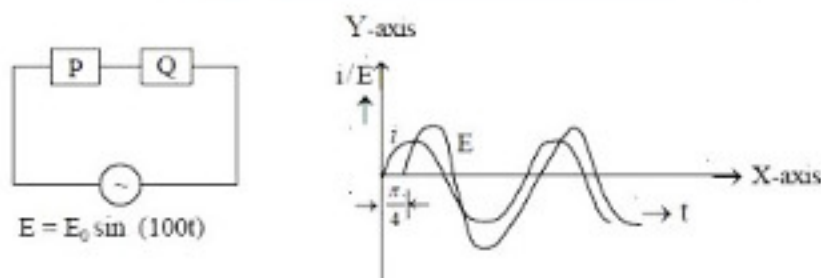
(4) 0.001 T

113. A conducting rod PQ of length 1 m is moving with a uniform speed 2 ms^{-1} in a uniform magnetic field of 4T which is directed into the paper. A capacitor of capacity $10 \mu\text{F}$ is connected as shown in the figure. Then, the charges on the plates of the capacitor are



- (1) $q_A = +80 \mu\text{C}$, $q_B = -80 \mu\text{C}$
- (2) $q_A = -80 \mu\text{C}$, $q_B = +80 \mu\text{C}$
- (3) $q_A = +1.25 \mu\text{C}$, $q_B = 1.25 \mu\text{C}$
- (4) $q_A = -1.25 \mu\text{C}$, $q_B = +1.25 \mu\text{C}$

114. For the AC circuit shown below, phase difference between emf and current is $\frac{\pi}{4}$ radian as shown in the graph. If the impedance of the circuit is 1414Ω then the values of P and Q are



- (1) $1 \text{ k}\Omega$, $10 \mu\text{F}$
 - (2) $1 \text{ k}\Omega$, $1 \mu\text{F}$
 - (3) $1 \text{ k}\Omega$, 10 mH
 - (4) $1 \text{ k}\Omega$, 1 mH
115. In a plane electromagnetic wave, the electric field oscillates with a frequency $2 \times 10^{10} \text{ s}^{-1}$ and amplitude 40 Vm^{-1} , then the energy density due to electric field is _____
 $(\epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1})$
- (1) $1.52 \times 10^{-9} \text{ Jm}^{-3}$

(2) $2.54 \times 10^{-19} \text{ Jm}^{-3}$

(3) $3.54 \times 10^{-9} \text{ Jm}^{-3}$

(4) $4.56 \times 10^{-9} \text{ Jm}^{-3}$

116. Photons of frequencies equal to the frequencies of H_{β} and H_{∞} lines of hydrogen incident on a photosensitive plate whose threshold frequency is equal to the frequency of H_{α} line of hydrogen. The ratio of the maximum kinetic energies of the emitted electrons is

(1) 7:16

(2) 3:4

(3) 8:27

(4) 5:36

117. Hydrogen atom is in its n^{th} energy state. If de-Broglie wavelength of the electron is λ , then

(1) $\lambda \propto \frac{1}{n^2}$

(2) $\lambda \propto \frac{1}{n}$

(3) $\lambda \propto n^2$

(4) $\lambda \propto n$

118. If 200 MeV of energy is released in the fission of one nucleus of ${}_{92}^{235}\text{U}$, the number of nuclei that must undergo fission to release an energy of 1000 J is

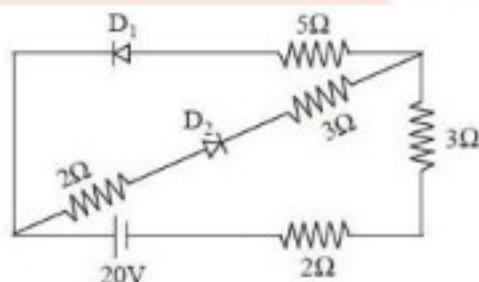
(1) 3.125×10^{13}

(2) 6.25×10^{13}

(3) 12.5×10^{13}

(4) 3.125×10^{14}

119. If the diodes are ideal in the circuit given below, then the current through the cell is



(1) 4 A

(2) 1.5 A

(3) 2 A

(4) 3 A

120. If a message signal of frequency 10 kHz and peak voltage 12 V is used to modulate a carrier wave of frequency 1 MHz. the modulation index is 0.6. To make the modulation index 0.75, the carrier peak voltage should be

(1) decreased by 25%

(2) increased by 25%

(3) decreased by 20%

(4) increased by 20%

Chemistry

121. If the radius of electron orbit in the excited state of hydrogen atom is 476.1 pm, the energy of the electron in that excited state in J is

(Radius and energy of electron in the first orbit of hydrogen atom are 52.9 pm and -2.18×10^{-8} J respectively)

- (1) -2.42×10^{-8}
- (2) -19.62×10^{-18}
- (3) -2.42×10^{-19}
- (4) -6.05×10^{-19}

122. A light of frequency 1.6×10^{16} Hz when falls on a metal plate emits electrons that have double the kinetic energy compared to the kinetic energy of emitted electrons when frequency of 1.0×10^{16} Hz falls on the same plate. The threshold frequency (ν_0) of the metal in Hz is

- (1) 1×10^{15}
- (2) 4×10^{15}

(3) 3×10^{15}

(4) 4×10^{13}

123. To which group and period does the element belong if the electronic configuration of an element in its -2 oxidation state is $1s^2 2s^2 2p^6 3s^2 3p^6$

(1) period 3, group 16

(2) period 3, group 17

(3) period 4, group 16

(4) period 4, group 17

124. Which set of the following molecules has only one lone pair of electrons on their respective central atoms?

a) SO_2 b) XeF_4 c) PbCl_2 d) SF_4 e) ClF_3

Identify the correct statements

(1) a, c, d

(2) b, c, d

(3) a, b, e

(4) a, c, e

125. XeF_4 is square planar where as CCl_4 is tetrahedral because

- (1) In XeF_4 , 'Xe' is sp^2 hybridized and in CCl_4 'C' is sp^3 hybridised
- (2) In both XeF_4 and CCl_4 the central atom is sp^3 hybridised
- (3) In XeF_4 , 'Xe' is sp^3d^2 hybridized but due to the presence of 2 lone pairs of electrons shape is square planar whereas in CCl_4 'C' is sp^3 hybridised
- (4) Xe is noble gas, whereas C is a non-metal.

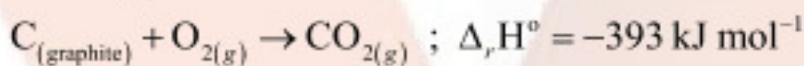
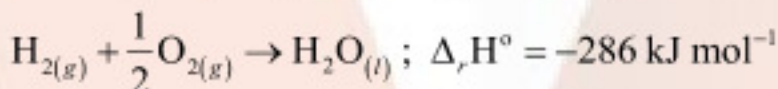
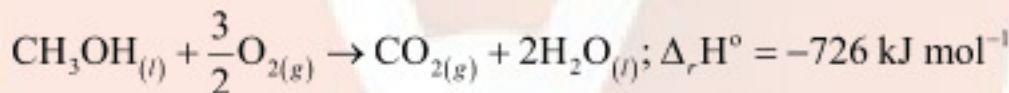
126. 16 g each of H_2 , He and O_2 are present in a container exerting 10 atm. pressure at T(K). The pressure in atm exerted by 16 g each of He and O_2 in the second container of same volume and temperature is

- (1) 1.8
- (2) 6.4
- (3) 3.6
- (4) 5.4

127. One litre of 0.15M Na_2SO_3 aqueous solution is mixed with 500 mL of 0.2 M $\text{K}_2\text{Cr}_2\text{O}_7$ aqueous solution in acid medium. What is the number of moles of $\text{K}_2\text{Cr}_2\text{O}_7$ remaining in the solution after the reaction?

- (1) 0.1
- (2) 0.0125
- (3) 0.025
- (4) 0.05

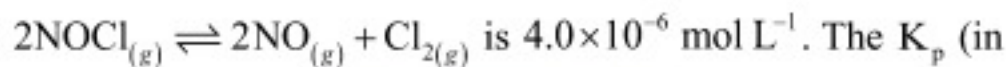
128. From the following data



The standard enthalpy of formation of $\text{CH}_3\text{OH}_{(l)}$ in kJ mol^{-1} is

- (1) -239
- (2) 239
- (3) 547
- (4) -905

129. At 1000 K, the equilibrium constant, K_c for the reaction



bar) at the same temperature is

$$(R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1})$$

(1) 3.32×10^{-6}

(2) 3.32×10^4

(3) 3.32×10^{-4}

(4) 3.32×10^{-3}

130. If the pK_a of acetic acid and pK_b of dimethylamine are 4.76 and 3.26 respectively, the pH of dimethyl ammonium acetate solution is

(1) 7.75

(2) 6.75

(3) 7.0

(4) 8.5

131. Which of the following statements are correct?

a) NaH(s) reacts with violently with water to form NaOH and H_2

b) An example for electron rich hydride is NH_3

c) Nickel; forms saline hydride

(1) a, c

(2) b, c

(3) a, b, c

(4) a, b

132. Which of the following nitrates on heating does **not** give its oxide?

(1) LiNO_3

(2) NaNO_3

(3) $\text{Ba}(\text{NO}_3)_2$

(4) $\text{Be}(\text{NO}_3)_2$

133. BF_3 reacts with NaH at 450 K to form NaF and X. When X reacts with LiH in diethyl ether, Y is formed. What is Y?

(1) LiBO_2

- (2) $\text{Li}_2\text{B}_4\text{O}_7$
- (3) LiBH_4
- (4) $\text{B}_2\text{H}_6 \cdot \text{LiH}$

134. Assertion (A) : $[\text{SiF}_6]^{2-}$ is formed but $[\text{SiCl}_6]^{2-}$ is not

Reason (R) : Electronegativity (EN) of F is higher than EN of Cl

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

135. The environmental friendly chemical now-a-days used for bleaching the paper in the presence of a suitable catalyst is

- (1) Chlorine
- (2) Sulphur dioxide
- (3) Hydrogen peroxide
- (4) Bleaching powder

136. The IUPAC name of the following compound is



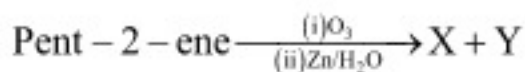
- (1) 5-Cyanopentan-2-one
- (2) 5-Oxohexanenitrile
- (3) 4-Oxopentanenitrile
- (4) 2-Oxopentanenitrile

137. Identify the correct statements from the following

- a. Petrol and CNG operated automobiles causes less pollution.
- b. Alkanes having tertiary hydrogen can be oxidized to alcohols by KMnO_4
- c. Methane can be prepared by Kolbe's electrolytic method
- d. Alkyl chloride on reduction with zinc and dilute hydrochloric acid gives alkane

- (1) b, c, d
- (2) a, b
- (3) a, b, d
- (4) c, d

138. What are X and Y in the following reaction?

XY

- (1) CH_3CHO $\text{CH}_3\text{CH}_2\text{CHO}$
(2) $\text{CH}_3\text{CH}_2\text{CHO}$ $\text{CH}_3\text{CH}_2\text{CHO}$
(3) CH_3CHO $(\text{CH}_3)_2\text{CO}$
(4) CH_3CHO CH_3CHO

139. The total number of body centered lattices possible among the 14 Bravais lattices is

- (1) 2
(2) 1
(3) 4
(4) 3

140. The measured osmotic pressure of a solution prepared by dissolving 17.4 mg of K_2SO_4 in 2L of water at 27°C is

3.735×10^{-3} bar. The Van't Hoff factor is

($R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$; atomic weights $\text{K} = 39$; $\text{S} = 32$; $\text{O} = 16$)

- (1) 2.84
- (2) 3.0
- (3) 2.0
- (4) 2.32

141. Dissolving 120 g of a compound (mol.wt = 60) in 1000 g of water gave a solution of density 1.12 g mL^{-1} . The molarity of solution is

- (1) 1.0 M
- (2) 2.0 M
- (3) 2.5 M
- (4) 4.0 M

142. When an aqueous solution of CuCl_2 is electrolysed using Pt electrodes, the reaction at cathode and anode respectively are

- (1) $4\text{H}_2\text{O}_{(l)} \xrightarrow{+4e^-} 2\text{H}_{2(g)} + 4\text{OH}^-_{(aq)} ; 2\text{H}_2\text{O}_{(l)} \xrightarrow{-4e^-} \text{O}_{2(g)} + 4\text{H}^+_{(aq)}$
- (2) $2\text{Cu}^{2+}_{(aq)} \xrightarrow{+4e^-} 2\text{Cu}_{(s)} ; 2\text{H}_2\text{O}_{(l)} \xrightarrow{-4e^-} \text{O}_{2(g)} + 4\text{H}^+_{(aq)}$
- (3) $\text{Cu}^{2+}_{(aq)} \xrightarrow{+2e^-} \text{Cu}_{(s)} ; 2\text{Cl}^-_{(aq)} \xrightarrow{-2e^-} \text{Cl}_{2(g)}$
- (4) $2\text{H}_2\text{O}_{(l)} \xrightarrow{+2e^-} \text{H}_{2(g)} + 2\text{OH}^-_{(aq)} ; 2\text{Cl}^-_{(aq)} \xrightarrow{-2e^-} \text{Cl}_{2(g)}$

143. Thermal decomposition of HCOOH is a first order reaction and the rate constant at $T(\text{K})$ is $4.606 \times 10^{-3} \text{ s}^{-1}$. The time required to decompose 90% of initial quantity of HCOOH at $T(\text{K})$ in seconds is

- (1) 100
- (2) 500
- (3) 1000
- (4) 50

144. Which one of the following statements is **not** correct?

- (1) A mixture of dinitrogen and dioxygen at room temperature is an example for aerosol.
- (2) Lyophilic sols are more stable compared to lyophobic sols
- (3) Formation of micelles is possible only above Kraft temperature
- (4) An example for a soap is sodium stearate and an example for detergent is sodium lauryl sulphate

145. In Ellingham diagram, the plot is drawn between

- (1) Temperature, ΔH°
- (2) Temperature, ΔG°
- (3) Pressure, ΔS°
- (4) Temperature, ΔE°

146. Identify the reaction which does **not** liberate N_2

- (1) $NaN_3 \xrightarrow{\Delta} ?$
- (2) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} ?$
- (3) $NH_4Cl + Ca(OH)_2 \rightarrow ?$
- (4) $Ba(N_3)_2 \xrightarrow{\Delta} ?$

147. Identify the molecule which contains lone pair of electrons on the sulphur atom

- (1) H_2SO_5
- (2) $H_2S_2O_8$
- (3) $H_2S_2O_7$
- (4) H_2SO_3

148. Which statement about noble gases is **not** correct?

- (1) 'Xe' forms XeF_6 under suitable conditions
- (2) 'Ar' is used in electric bulbs.
- (3) The number of lone pair of electrons present on Xe in XeF_2 is 3.
- (4) 'He' has the highest boiling point among all the noble gases

149. Crystal field splitting energies for octahedral (Δ_o) and tetrahedral (Δ_t) geometries caused by the same ligands are related through the expression

- (1) $\Delta_o = \Delta_t$
- (2) $4\Delta_o = 9\Delta_t$
- (3) $9\Delta_o = 4\Delta_t$
- (4) $\Delta_o = 2\Delta_t$

150. In Lanthanide series, the element well known to exhibit +4 oxidation state is

- (1) Lu
- (2) Ce

(3) Pm

(4) Nd

151. In anionic polymerization, the compound which acts as effective chain initiator is

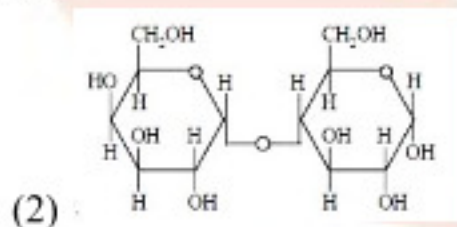
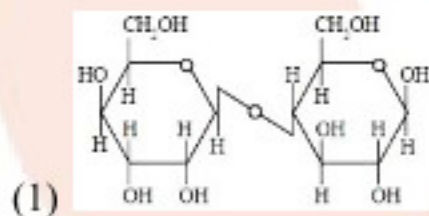
(1) BF_3

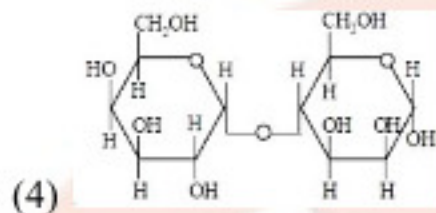
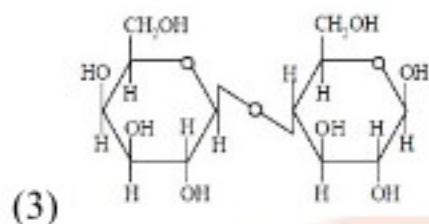
(2) $(\text{CH}_3\text{CO})_2\text{O}_2$

(3) SnCl_2

(4) R-Li

152. Which one of the following is the structure of lactose?



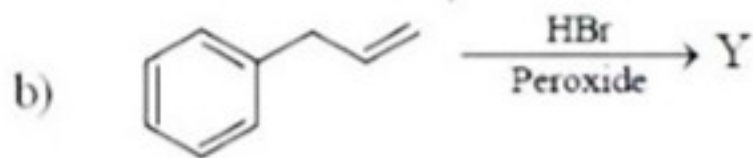


153. Which of the following statements are correct?

- a) Drugs that mimic natural messenger by switching on the receptor are called agonists
- b) Shape of the receptor does not change after attachment of chemical messenger
- c) A cationic detergent is formed when stearic acid reacts with polyethylene glycol
- d) Seldane is an antihistamine

- (1) b, c
- (2) a, c, d
- (3) a, d
- (4) a, b, c

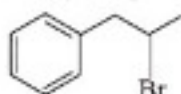
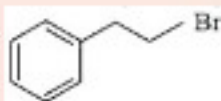
154. Identify the major products X and Y in the following reactions



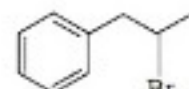
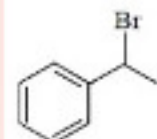
X

Y

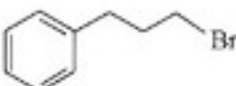
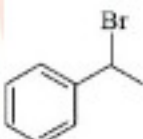
1.



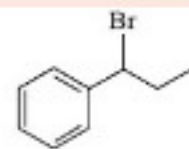
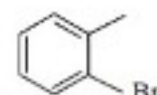
2.



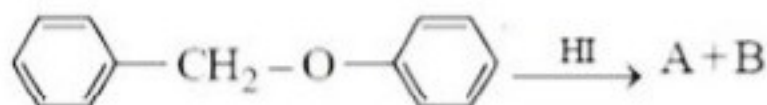
3.

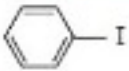
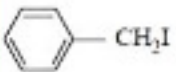
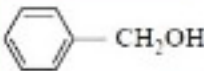


4.

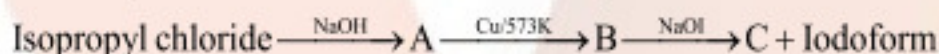


155. Identify A and B in the following reactions?

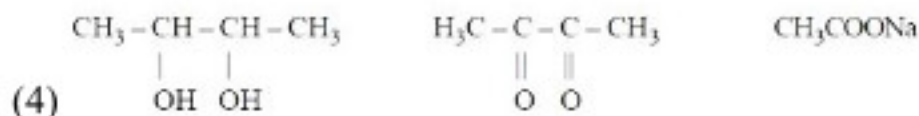


- | | <u>A</u> | <u>B</u> |
|-----|---|---|
| (1) |  | $\text{H}_3\text{C}-\text{O}-\text{C}_6\text{H}_5$ |
| (2) |  | $\text{HO}-\text{C}_6\text{H}_5$ |
| (3) |  | $\text{I}-\text{C}_6\text{H}_5$ |
| (4) |  | $\text{IH}_2\text{C}-\text{O}-\text{C}_6\text{H}_5$ |

156. Identify A, B and C in the following reactions



- | | A | B | C |
|-----|--|---|--------------------------------------|
| (1) | $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ | $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ | $\text{CH}_3\text{CH}_2\text{COONa}$ |
| (2) | $\text{CH}_3\text{CH}_2\text{OH}$ | CH_3CHO | HCOONa |
| (3) | $\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_3 \\ \\ \text{OH} \end{array}$ | CH_3COCH_3 | CH_3COONa |



157. Match the following

List-I	List-II
A) Lucas reagent	I) $\text{SnCl}_2 + \text{HCl}, \text{H}_3\text{O}^+$
B) Clemmensen reagent	II) $[\text{Ag}(\text{NH}_3)_2]^+$
C) Tollen's reagent	III) Anhydrous ZnCl_2 conc. HCl
D) Stephen reaction	IV) $\text{Zn} - \text{Hg}$ conc. HCl
	V) $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$

The correct answer is

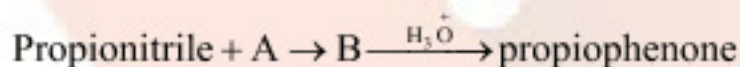
- | | | | | |
|-----|-----|-----|-----|----|
| | A | B | C | D |
| (1) | III | IV | II | I |
| | A | B | C | D |
| (2) | III | IV | I | II |
| | A | B | C | D |
| (3) | IV | II | III | V |
| | A | B | C | D |
| (4) | IV | III | I | V |

158. What are A, B and C in the following reactions?

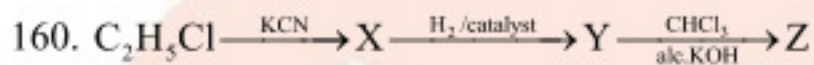


- | <u>A</u> | <u>B</u> | <u>C</u> |
|----------|----------|----------|
| (1) | | |
| (2) | | |
| (3) | | |
| (4) | | |

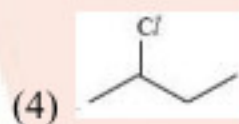
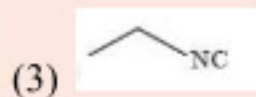
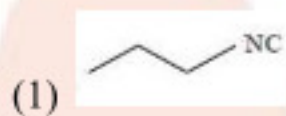
159. What are A and B in the following reaction sequence?



- | <u>A</u> | <u>B</u> |
|---------------------------------------|---|
| (1) $\text{C}_2\text{H}_5\text{MgBr}$ | $\begin{array}{c} \text{NMgBr} \\ \\ \text{CH}_3\text{CH}_2 - \text{C} - \text{C}_2\text{H}_5 \end{array}$ |
| (2) $\text{C}_2\text{H}_5\text{MgBr}$ | $\begin{array}{c} \text{NH} \\ \\ \text{CH}_3\text{CH}_2\text{CH}_2 - \text{C} - \text{C}_2\text{H}_5 \end{array}$ |



What is 'Z' in the above sequence of reactions?



AP EAMCET-2018-Engineering-22 Apr-Shift-1
Answer Key

1	4	33	2	65	4	97	4	129	3
2	4	34	2	66	4	98	2	130	1
3	4	35	1	67	2	99	1	131	4
4	3	36	*	68	*	100	3	132	2
5	4	37	2	69	4	101	3	133	3
6	1	38	2	70	4	102	1	134	2
7	4	39	2	71	2	103	1	135	3
8	1	40	2	72	3	104	4	136	2
9	4	41	4	73	3	105	1	137	3
10	4	42	3	74	4	106	4	138	1
11	2	43	4	75	2	107	1	139	4
12	1	44	2	76	4	108	2	140	2
13	1	45	2	77	4	109	3	141	2
14	1	46	3	78	1	110	4	142	3
15	2	47	3	79	3	111	1	143	2
16	2	48	3	80	4	112	3	144	1
17	3	49	4	81	1	113	1	145	2

18	3	50	3	82	4	114	1	146	3
19	4	51	2	83	3	115	3	147	4
20	2	52	4	84	4	116	1	148	4
21	4	53	2	85	4	117	4	149	2
22	3	54	1	86	2	118	1	150	2
23	2	55	3	87	1	119	3	151	4
24	4	56	1	88	1	120	3	152	3
25	1	57	3	89	4	121	3	153	3
26	3	58	2	90	2	122	2	154	3
27	2	59	2	91	1	123	1	155	2
28	4	60	1	92	4	124	1	156	3
29	2	61	4	93	1	125	3	157	1
30	4	62	2	94	3	126	3	158	4
31	1	63	2	95	2	127	4	159	3
32	2	64	1	96	3	128	1	160	1

AP EAMCET-2018-Engineering-22 Apr-Shift-1
Solutions – Mathematics

1. The function is given as,

$$f(x) = [2x] - 2[x] \text{ for } x \in \mathbb{R}$$

The x is comes out as,

$$x = [x] + \{x\}$$

$$2x = 2[x] + 2\{x\}$$

$$[2x] = 2[x] + (2\{x\})$$

The range for function $2x$ is,

$$[2x] = \begin{cases} 2[x] + 0, & 0 < \{x\} < \frac{1}{2} \\ 2[x] + 1, & \frac{1}{2} \leq \{x\} < 1 \end{cases}$$

Putting the values in given function.

$$[2x] - 2[x] = \begin{cases} 0, & 0 \leq \{x\} < \frac{1}{2} \\ 1, & \frac{1}{2} \leq \{x\} < 1 \end{cases}$$

Thus, the range of the function is $\{0, 1\}$.

2. The function is given as,

$$f(x) = \log\{ax^3 + (a+b)x^2 + (b+c)x + c\}$$

$$f(x) = \log\{ax^2 + bx + c(x+1)\}$$

To define the function its value must be greater than 0.

$$(ax^2 + bx + c)(x+1) > 0$$

$$x+1 > 0 \text{ and } x \neq -\frac{b}{2a}$$

Thus, the domain becomes,

$$D = R - \left\{-\frac{b}{2a}\right\} \cup (-\infty, -1]$$

3. The given function is,

$$(15 \times 5^{2n}) + (2 \times 2^{3n})$$

At $n = 1$

$$(15 \times 5^2) + (2 \times 2^3) = 391$$

The number 391 gets divisible by 17.

Thus, $(15 \times 5^{2n}) + (2 \times 2^{3n})$ is divisible by 17.

4. The determinant of A comes out to be,

$$|A| = \begin{vmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{vmatrix}$$

$$|A| = 3(-3 + 4) + 3(2) + 4(-2) = 1 \neq 0$$

Taking the adjacent of A we get,

$$\text{adj}(A) = \begin{bmatrix} 1 & -2 & -2 \\ -1 & 3 & 3 \\ 0 & -4 & -3 \end{bmatrix}^t = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

$$A^{-1} = \frac{\text{adj}(A)}{|A|} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

On squaring the given determinant,

$$A^2 = A \cdot A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix} \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

Thus, the inverse of A comes out to be $A^{-1} = A^3$.

5. The given matrices are,

$$A = \begin{bmatrix} k/2 & 0 & 0 \\ 0 & l/3 & 0 \\ 0 & 0 & m/4 \end{bmatrix} \text{ and } A^{-1} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/4 \end{bmatrix}$$

From property of matrices $AA^{-1} = I$

Here, the matrices I is identity matrices.

$$\therefore \begin{bmatrix} k/2 & 0 & 0 \\ 0 & l/3 & 0 \\ 0 & 0 & m/4 \end{bmatrix} \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

On solving we get,

$$k = 4 \times 1 = 4$$

$$l = 9 \times 1 = 9$$

$$m = 16 \times 1 = 16$$

$$\therefore k + l + m = 4 + 9 + 16 = 29$$

6. The determinant form of given equation is,

$$D = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 3 & 4 \\ 1 & 5 & 10 \end{vmatrix}$$

$$D = 1(30 - 20) - 2(10 - 4) + 1(5 - 3)$$

$$D = 0$$

This shows the given equation of determinant is consistent.

$$\therefore D_1 = 0$$

$$D_1 = \begin{vmatrix} 1 & 2 & 1 \\ k & 3 & 4 \\ k^2 & 5 & 10 \end{vmatrix}$$

$$D_1 = 1(30 - 20) - 2(10k - 4k^2) + (5k - 3k^2) = 0$$

$$5k^2 - 15k + 10 = 0$$

$$k = 2, 1$$

Thus, the value of $A = 2$ and $B = 1$

$$\therefore A + B = 2 + 1 = 3$$

7. The given function is,

$$\frac{z-1}{z+i} = \frac{x+iy-1}{x+iy+i} = \frac{(x-1)+iy}{x+(y+1)i} \times \frac{x-(y+1)i}{x-(y+1)i}$$

$$\frac{z-1}{z+i} = \frac{x(x-1)+y(y+1)}{x^2+(y+1)^2} + \frac{[xy-(x-1)(y+1)]i}{x^2+(y+1)^2}$$

It is given $\operatorname{Re}\left(\frac{z-1}{z+i}\right) = 1$

$$x(x-1)+y(y+1)=x^2+(y+1)^2$$

$$-x+y=2y+1$$

$$x+y+1=0$$

Thus, the point $(2016, -2017)$ lies on line $x+y+1=0$

8. It is given that,

$$13e^{i \tan^{-1} \frac{5}{12}} = a+ib$$

On expanding the exponential value we get,

$$13 \left[\cos \left(\tan^{-1} \frac{5}{12} \right) + i \sin \left(\tan^{-1} \frac{5}{12} \right) \right] = a + ib$$

$$13 \left[\cos \left(\cos^{-1} \frac{12}{13} \right) + i \sin \left(\sin^{-1} \frac{5}{13} \right) \right] = a + ib$$

$$13 \left[\frac{12}{13} + i \frac{5}{13} \right] = a + ib$$

$$a = 12, b = 5$$

Therefore the pair becomes (12, 5).

9. It is given that,

$$z_1 = 1 - 2i, z_2 = 1 + i$$

$$z_3 = 3 + 4i$$

Now,

$$\left(\frac{1}{z_1} + \frac{3}{z_2} \right) \frac{z_3}{z_2} = \left(\frac{1}{1-2i} + \frac{3}{1+i} \right) \left(\frac{3+4i}{1+i} \right)$$

$$\left(\frac{1}{z_1} + \frac{3}{z_2} \right) \frac{z_3}{z_2} = \frac{64 + 2i - 32i + 1}{2(4+1)} = \frac{65 - 30i}{10} = \frac{13}{2} - 3i$$

10. The given equation is,

$$\begin{aligned}\frac{1}{1+2\omega} + \frac{1}{2+\omega} - \frac{1}{1+\omega} &= \frac{(3+3\omega)(1+\omega) - (1+2\omega)(2+\omega)}{(1+2\omega)(2+\omega)(1+\omega)} \\ &= \frac{3+6\omega+3\omega^2-2-\omega-4\omega-2\omega^2}{(1+2\omega)(2+\omega)(1+\omega)} = \frac{1+\omega+\omega^2}{(1+2\omega)(2+\omega)(1+\omega)}\end{aligned}$$

Substitute $1+\omega+\omega^2=0$ in above expression.

$$\frac{1}{1+2\omega} + \frac{1}{2+\omega} - \frac{1}{1+\omega} = 0$$

11. We have, $5x-2 < (x+1)^2$

On solving the equation we get,

$$5x-1 < x^2+2x+1$$

$$x^2-3x+2 > 0$$

$$x \in (-\infty, 1) \cup (2, \infty)$$

Again,

$$(x+1)^2 < 7x-3$$

$$x^2+2x+1 < 7x-3$$

$$x^2-5x+4 < 0$$

$$x \in (1, 4)$$

Thus, the number of integral satisfying equation is 3.

12. The given function of $f(x)$ is,

$$f(x) = x^2 + 2bx + 2c^2$$

$$f(x) = (x+b)^2 + 2c^2 - b^2$$

The minimum value of $f(x) = 2c^2 - b^2$

The given function of $g(x)$ is,

$$g(x) = -x^2 - 2cx + b^2 = -\left[(x+c)^2 - b^2 - c^2\right]$$

$$g(x) = -(x+c)^2 + b^2 + c^2$$

The maximum value of $g(x) = b^2 + c^2$

According to given conditions,

$$2c^2 - b^2 > b^2 + c^2$$

$$c^2 > 2b^2$$

13. The equation is given as,

$$x^3 + qx + r = 0$$

The roots of the given equation is,

$$a + b + c = 0$$

$$ab + bc + ca = q$$

$$abc = -r$$

Squaring the root of the equation.

$$(a + b + c)^2 = 0$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ca = 0$$

$$a^2 + b^2 + c^2 + 2(ab + bc + ca) = 0$$

Substituting the values we get,

$$a^2 + b^2 + c^2 = -2q$$

Now,

$$(a - b)^2 + (b - c)^2 + (c - a)^2$$

$$= a^2 + b^2 - 2ab + b^2 + c^2 - 2bc + c^2 + a^2 - 2ac$$

$$= 2(a^2 + b^2 + c^2) - 2(ab + bc + ca)$$

$$= 9(-2q) - 2(q)$$

$$= -6q$$

14. It is assumed the roots of the equations are $m, -m$

The sum of three roots $= 2p$

Hence, third root will be $2p$

Now,

$$m \times (-m) + m(2p) + (-m) \times 2p = 3q$$

$$-m^2 = 3q$$

Now,

$$m \times (-m) \times 2p = 4r$$

$$3q \times 2p = 4r$$

$$r = \frac{3pq \times 2}{4} = \frac{3pq}{2}$$

15. The four even digits are 0, 3, 5, 4

The digits without reception are

3054, 3504, 5034, 5304, 3450, 3540, 4350, 4530, 5340, 5430

The required sum becomes,

$$N = 3054 + 3504 + 5034 + 5304 + 3450 + 3540 + 4350 + 4530 + 5340 + 5430$$

$$N = 43536$$

16. Boys can be seated in a row in 6P_6 ways = 6!

The probability to arrange 6 girls in 7 gaps is,

$$x = {}^7P_6 \times 6!$$

$$x = 7 \times 6!$$

The number of ways 6 boys can be seated in circle is,

$$N = (6-1)! = 5!$$

Now, the 6 girls are arranged in 6 gaps in 6P_6 ways.

$$y = {}^6P_6 \times 5!$$

$$y = 6! \times 5!$$

The ratio of $x : y$ becomes,

$$x : y = 6! \times 7! : 5! \times 6!$$

$$x : y = 7! \times 5!$$

$$x : y = 7 \times 6 \times 5! : 5! = 42 : 1$$

17. In 7 letter word SARANAM, the letter A comes 3 times and all other are distinct.

The total five letter words consist of,

Total words = all letters using SARNM + using 2 A and 3 other
+ using 3 A and 2 other

$$\text{Total words} = 5! + {}^4C_3 \times \frac{5!}{2!} + {}^4C_2 \times \frac{5!}{3!}$$

$$\text{Total words} = 120 + 4 \times \frac{120}{2} + 6 \times \frac{120}{6} = 480$$

18. On expanding the given expression we get,

$$\left(\sqrt[4]{5} + \sqrt[5]{4}\right)^{100} = \sum_{r=0}^{100} {}^{100}C_r \left(\sqrt[4]{5}\right)^{100-r} \left(\sqrt[5]{4}\right)^r$$

$$\left(\sqrt[4]{5} + \sqrt[5]{4}\right)^{100} = \sum_{r=0}^{100} {}^{100}C_r 5^{\frac{100-r}{4}} 4^{r/5} = \sum_{r=0}^{100} T_{r+1}$$

Here, $T_{r+1} = {}^{100}C_r 5^{\frac{100-r}{4}} 4^{r/5}$

Clearly, T_{r+1} will be an integer if $\frac{100-r}{4}$ and $\frac{r}{5}$ are integers.

This is possible when $100-r$ is a multiple of 4 and r is multiple of 5.

When $100-r$ is a multiple of (4)

$$100-r = 0, 4, 8, 12, \dots, 96, 100$$

When $100-r$ is a multiple of 5.

$$100-r = 0, 5, 10, \dots, 100$$

$$\therefore r = 0, 4, 8, 12, \dots, 96, 100 \text{ and,}$$

$$r = 0, 5, 10, 100$$

The rational number terms comes out to be,

$$r = 0, 20, 40, 60, 80, 100$$

Thus, there are 6 rational terms.

19. On expanding the given expression,

$$(2a - 3b)^{19} = 2^{19} \cdot a^{19} \left(1 - \frac{3b}{2a}\right)^{19}$$

The r^{th} term of greatest term is,

$$(1+x)^n = \left[\frac{(n+1)|x|}{1+|x|} \right]$$

Here, $n = 19$, $x = -\frac{3b}{2a}$

$$\therefore r = \left[\frac{20 \left| \frac{3b}{2a} \right|}{1 + \frac{3b}{2a}} \right] = \left[\frac{20}{1+4} \times 4 \right]$$

$$r = 16$$

The greatest term of $(2a - 3b)^{19}$ is,

$$(2a - 3b)^{19} = {}^{19}C_{16} \times 2^{19} \cdot a^{19} \left(\frac{3b}{2a} \right)^{16}$$

$$(2a - 3b)^{19} = {}^{19}C_{16} \times 2^{19} \cdot \left(\frac{1}{4} \right)^{19} (4)^{16} = {}^{19}C_3 \times 2^{13}$$

20. It is given,

$$\frac{x^2 + 5x + 7}{(x-3)^3} = \frac{A}{x-3} + \frac{B}{(x-3)^2} + \frac{C}{(x-3)^3}$$

$$x^2 + 5x + 7 = A(x-3)^2 + B(x-3) + C \quad \dots\dots (I)$$

Substituting the value of x as 3 in above equation we get,

$$C = 9 + 15 + 7 = 31$$

Substituting the value of x as 0 in above equation we get,

$$9A - 3B + 31 = 7$$

$$3A - B = -8$$

Substituting the value of x as 1 in above equation we get,

$$4A - 2B + 31 = 1 + 5 + 7$$

$$2A - B = -9$$

On solving the above two to obtain the value of A and B ,

$$A = 1 \text{ and } B = 11$$

The equation of line having slope A and passing through the point (B, C) is,

$$y - 31 = 1(x - 11)$$

$$y - 31 = x - 11$$

$$x - y + 20 = 0$$

21. It is given that,

$$\cos\left(x - \frac{\pi}{3}\right), \cos x, \cos\left(x + \frac{\pi}{3}\right) \text{ are in H.P.}$$

$$\cos x = \frac{2 \cos\left(x - \frac{\pi}{3}\right) \cos\left(x + \frac{\pi}{3}\right)}{\cos\left(x - \frac{\pi}{3}\right) + \cos\left(x + \frac{\pi}{3}\right)}$$

$$\cos x = \frac{2\left(\cos^2 x - \sin^2 \frac{\pi}{3}\right)}{2 \cos x \times \cos \frac{\pi}{3}}$$

$$\cos^2 x \left(1 - \cos \frac{\pi}{3}\right) = \sin^2 \frac{\pi}{3}$$

$$\cos^2 x \left(1 - \cos \frac{\pi}{3}\right) = \left(1^2 - \cos^2 \frac{\pi}{3}\right)$$

Expanding the above we get,

$$\cos^2 x \left(1 - \cos \frac{\pi}{3}\right) = \left(1 - \cos \frac{\pi}{3}\right) \left(1 + \cos \frac{\pi}{3}\right)$$

$$\cos^2 x = 1 + \cos \frac{\pi}{3}$$

$$\cos^2 x = 2 \cos^2 \frac{\pi}{6} = 2 \times \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{2}$$

$$\cos x = \sqrt{\frac{3}{2}}$$

22. It is given that,

$$\cos^3 10^\circ + \cos^3 110^\circ + \cos^3 130^\circ$$

We know,

$$\cos^3 x + \cos^3 (120 - x) + \cos^3 (120 + x) = \frac{3}{4} \cos 3x$$

In this case x is taken as 10 .

$$\begin{aligned} \cos^3 10^\circ + \cos^3 110^\circ + \cos^3 130^\circ &= \cos^3 10^\circ + \\ &\quad \cos^3 (120^\circ - 10^\circ) + \cos^3 (120^\circ + 10^\circ) \\ \cos^3 10^\circ + \cos^3 110^\circ + \cos^3 130^\circ &= \left(\frac{3}{4}\right) \cos(3 \times 10^\circ) \\ &= \frac{3}{4} \cos 30^\circ = \frac{3\sqrt{3}}{8} \end{aligned}$$

23. It is given,

$$\sin 5x = \cos 2x$$

$$\sin 5x = \sin \left(\frac{\pi}{2} - 2x \right)$$

On simplifying the above we get,

$$5x = n\pi + (-1)^n \left(\frac{\pi}{2} - 2x \right)$$

$$5x = n\pi + (-1)^n \frac{\pi}{2} - (-1)^n 2x$$

$$x(5 + (-1)^n 2) = \frac{\pi}{2} (2n + (-1)^n)$$

$$x = \frac{\pi}{2} \left(\frac{2n + (-1)^n}{5 + 2(-1)^n} \right) = \frac{\pi}{2} \cdot a_n$$

Thus,

$$a_n = \frac{2n + (-1)^n}{5 + 2(-1)^n}$$

24. It is assumed $x = \tan \theta$, then we have

$$a \tan \theta + b \sec \theta = c$$

$$a \sin \theta + b = c \cos \theta$$

$$c \cos \theta - a \sin \theta = b$$

$$\text{Let, } a^2 + c^2 = r^2$$

$$c = r \cos \alpha, a = r \sin \alpha$$

$$\tan \alpha = \frac{a}{c}$$

$$\cos \alpha \times \cos \theta - \sin \alpha \times \sin \theta = \frac{b}{r}$$

$$\text{Thus, } \cos(\alpha + \theta) = \frac{b}{r}$$

$$\alpha + \theta = \pm \cos^{-1} \frac{b}{r}$$

Let, $\alpha + \theta$ be the positive solution and $\alpha + \phi$ the negative solution, where

$$y = \tan \phi$$

$$\alpha + \phi = -(\alpha + \theta)$$

$$-2\alpha = \theta + \phi$$

$$\tan(-2\alpha) = \tan(\theta + \phi)$$

On expanding the above expression we get,

$$\frac{-2 \tan \alpha}{1 - \tan^2 \alpha} = \tan(\theta + \phi)$$

Substitute the values in above expression.

$$\frac{-2a/c}{1 - (a/c)^2} = \frac{x+y}{1-xy}$$

$$\frac{2ac}{a^2 - c^2} = \frac{x+y}{1-xy}$$

25. The value of

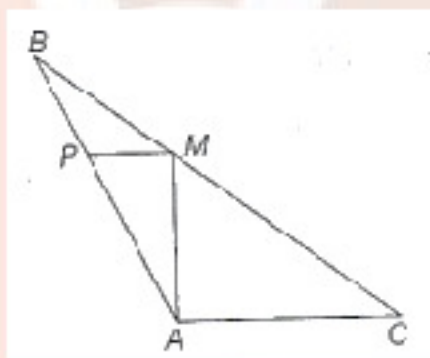
$$\tanh^{-1} \frac{1}{2} + \coth^{-1} 3 = \frac{1}{2} \ln \left(\frac{1 + \frac{1}{2}}{1 - \frac{1}{2}} \right) + \frac{1}{2} \ln \left(\frac{3+1}{3-1} \right) = \frac{1}{2} \log \left(\frac{4}{2} \right)$$

$$\tanh^{-1} \frac{1}{2} + \coth^{-1} 3 = \frac{1}{2} \log 3 + \frac{1}{2} \log 2$$

$$\tanh^{-1} \frac{1}{2} + \coth^{-1} 3 = \log \sqrt{3} + \log \sqrt{2}$$

$$\tanh^{-1} \frac{1}{2} + \coth^{-1} 3 = \log \sqrt{6}$$

26. Consider the figure shown below,



Here, AM is the median and AC and AM are perpendicular.

The length BM comes out to be,

$$BM = CM = 2BC$$

From the property of triangle,

$$\angle A + \angle B + \angle C = 180^\circ$$

In $\triangle AMC$,

$$\angle AMC + \angle CAM + \angle MCA = 180^\circ$$

$$\angle AMC + 90^\circ + \angle C = 180^\circ$$

$$\angle AMC = 90^\circ - \angle C$$

Again at point M ,

$$\angle BMP + \angle PMA + \angle AMC = 180^\circ$$

$$\angle BMP + 90^\circ + 90^\circ - \angle C = 180^\circ$$

$$\angle BMP = \angle C$$

In $\triangle BPM$,

$$\angle BPM + \angle BMP + \angle PBM = 180^\circ$$

$$\angle BPM + \angle C + \angle B = 180^\circ$$

Substituting the above values we get,

$$\angle BPM = \angle A$$

Now, it is evident that ABC and MPB are equivalent triangle,

$$\frac{BM}{BC} = \frac{BP}{AB} = \frac{PM}{AC} = \frac{1}{2}$$

Also,

$$\angle BAC = \angle BAM + \angle MAC$$

$$\angle BAM = \angle BAC - \angle MAC$$

$$\angle BAM = \angle A - 90^\circ$$

In $\triangle AMC$,

$$\angle PAM + \angle APM + \angle AMP = 180^\circ$$

$$\angle A - 90^\circ + \angle APM + 90^\circ = 180^\circ$$

$$\tan(\angle APM) = \frac{AM}{PM}$$

$$\tan(180^\circ - A) = \frac{AM}{PM}$$

$$-\tan A = \frac{AM}{PM}$$

$$\text{Now, in } \triangle ACM, \tan C = \frac{AM}{AC}$$

$$AM = AC \tan C = -PM \tan A$$

$$PM \tan A + AC \tan C = 0$$

$$\frac{MP}{AC} = \frac{1}{2}$$

$$AC = 2MP$$

Hence,

$$PM \tan A + 2PM \tan C = 0$$

$$PM (\tan A + 2 \tan C) = 0$$

$$\tan A + 2 \tan C = 0$$

$$\frac{\tan A}{\tan C} = -2$$

27. On expanding the given expression,

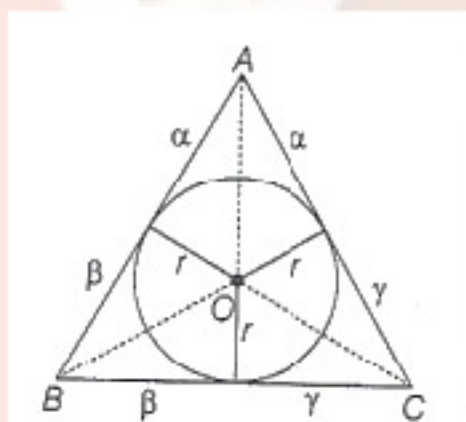
$$\left(\tan \frac{A}{2} + \tan \frac{B}{2} \right) = \frac{\Delta}{s(s-a)} + \frac{\Delta}{s(s-b)}$$

$$\left(\tan \frac{A}{2} + \tan \frac{B}{2} \right) = \frac{\Delta}{s} \left[\frac{1}{s-a} + \frac{1}{s-b} \right] = \frac{c\Delta}{s(s-a)(s-b)}$$

$$\left(\tan \frac{A}{2} + \tan \frac{B}{2} \right) = \frac{2c\Delta}{(a+b+c)(s-a)(s-b)}$$

$$\Rightarrow \frac{2c}{(a+b+c)} \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} = \frac{2c \cot \frac{C}{2}}{a+b+c}$$

28. Consider the figure shown below,



From figure,

$$\text{ar}(\triangle ABC) = \text{ar}(\triangle AOB) + \text{ar}(\triangle BOC) + \text{ar}(\triangle COA)$$

$$\text{Let, } S = \text{ar}(\triangle ABC)$$

$$S = \frac{1}{2}cr + \frac{1}{2}ar + \frac{1}{2}br$$

$$S = \frac{1}{2}r(a + b + c) = \frac{1}{2}r(2p) = rp$$

Here, the semi perimeter is p .

$$S = rp$$

$$r = \frac{S}{p}$$

Squaring the above expression we get,

$$r^2 = \frac{S^2}{p^2}$$

$$r^2 = \frac{S(S-a)(S-b)(S-c)}{p^2}$$

$$r^2 = \frac{\alpha \times \beta \times \gamma}{\alpha + \beta + \gamma} \quad [\because 2S = 2\alpha + 2\beta + 2\gamma]$$

29. The equation of line joining the points,

$$\vec{a} + 2\vec{b} - 5\vec{c}, -\vec{a} - 2\vec{b} - 3\vec{c} \text{ is,}$$

$$\vec{r} = (\vec{a} + 2\vec{b} - 5\vec{c}) + \lambda_1 (2\vec{a} - 4\vec{b} - 2\vec{c})$$

Similarly, equation of line joining the points

$$-4\vec{c}, 6\vec{a} - 4\vec{b} + 4\vec{c} \text{ is,}$$

$$\vec{r} = (-4\vec{c}) + \lambda_2 (6\vec{a} - 4\vec{b} + 8\vec{c})$$

Now, the point of intersection from above lines is,

$$\begin{aligned} & (2\lambda_1 + 1)\vec{a} + (4\lambda_1 + 2)\vec{b} + (-2\lambda_1 - 5)\vec{c} \\ &= (6\lambda_2)\vec{a} + (-4\lambda_2)\vec{b} + (8\lambda_2 + 4)\vec{c} \end{aligned}$$

On comparing we get,

$$2\lambda_1 + 1 = 6\lambda_2$$

$$4\lambda_1 + 2 = -4\lambda_2$$

$$-2\lambda_1 - 5 = 8\lambda_2 + 4$$

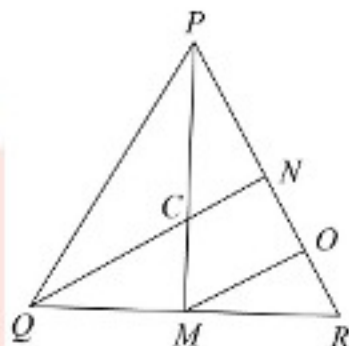
On solving the above three equation we get,

$$\lambda_1 = -\frac{1}{2} \text{ and } \lambda_2 = 0$$

Thus, the intersection points are $-4\vec{c}$ and for $\mu = -3$ the line

$\vec{r} = (3\vec{a} + 6\vec{b} - \vec{c}) + \mu(\vec{a} + 2\vec{b} + \vec{c})$ passes through the point $-4\vec{c}$ only.

30. Consider the figure shown below,



A line MO is drawn which is parallel to CN .

Since,

N is the midpoint of PM .

$$CN = \frac{1}{2}PM \dots\dots (I)$$

M is the midpoint of QR .

$\therefore MO$ is parallel to QN .

$$MO = \frac{1}{2}QN$$

Substitute the value in equation (I).

$$CN = \frac{1}{2} \left(\frac{1}{2}QN \right) = \frac{1}{4}QN$$

$$\frac{QN}{CN} = 4$$

31. It is given,

$$\vec{a} = \hat{i} - 2\hat{j} - 3\hat{k}$$

$$\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$$

$$\vec{c} = \hat{i} + 3\hat{j} - 2\hat{k}$$

Now,

$$(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c}) = [\vec{a} \vec{b} \vec{c}] \vec{b}$$

$$(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a}) = [\vec{a} \vec{b} \vec{c}] \vec{c}$$

$$(\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b}) = [\vec{a} \vec{b} \vec{c}] \vec{a}$$

Now,

$$\therefore [(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c})](\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})(\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b})]$$

$$= [-20\vec{b} - 20\vec{c} - 20\vec{a}]$$

$$= \begin{vmatrix} -40 & -20 & 20 \\ -20 & -60 & 40 \\ -20 & 40 & 60 \end{vmatrix} = -40(-3600 - 1600) + 20(-1200 + 800) + 20(-2000)$$

$$= 160000$$

32. It is given that,

$$\vec{n} \perp \vec{a} \text{ and } \vec{n} \perp \vec{b}$$

$$\therefore \vec{n} = \vec{a} \times \vec{b}$$

$$= \begin{vmatrix} i & j & k \\ 1 & 2 & 3 \\ -1 & 2 & 1 \end{vmatrix} = -4\hat{i} - 4\hat{j} + 4\hat{k}$$

$$\text{Again, } \sin \theta = \frac{|\vec{n} \times \vec{c}|}{|\vec{n}||\vec{c}|}$$

Now,

$$\vec{n} \times \vec{c}$$

$$= \begin{vmatrix} i & j & k \\ -4 & -4 & 4 \\ 1 & 2 & -2 \end{vmatrix} = -4\hat{j} - 4\hat{k}$$

$$\therefore \sin \theta = \frac{|-4\hat{j} - 4\hat{k}|}{|-4\hat{i} - 4\hat{j} + 4\hat{k}||\hat{i} + 2\hat{j} - 2\hat{k}|}$$

$$\sin \theta = \frac{\sqrt{(-4)^2 + (-4)^2}}{\sqrt{(-4)^2 + (-4)^2 + (4)^2} \times \sqrt{(1)^2 + (2)^2 + (-2)^2}}$$

$$\sin \theta = \frac{\sqrt{2}}{3\sqrt{3}}$$

33. It is assumed,

$$|\vec{a}| = |\vec{b}| = |\vec{c}| = \lambda$$

Now,

$$|\vec{a} + \vec{b} + \vec{c}|^2 = (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c})$$

$$|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 = 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

$$|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 = \lambda^2 + \lambda^2 + \lambda^2 + 2(0) = 3\lambda^2$$

$$|\vec{a} + \vec{b} + \vec{c}| = \sqrt{3}\lambda$$

Let the angle between $\vec{a}$ and $\vec{a} + \vec{b} + \vec{c}$ is θ .

$$\therefore \cos \theta = \frac{\vec{a}(\vec{a} + \vec{b} + \vec{c})}{|\vec{a}||\vec{a} + \vec{b} + \vec{c}|} = \frac{|\vec{a}|^2}{|\vec{a}||\vec{a} + \vec{b} + \vec{c}|}$$

$$\cos \theta = \frac{\lambda^2}{\lambda + \sqrt{3}\lambda} = \frac{1}{\sqrt{3}}$$

34. Let,

$$\begin{aligned} A(2\vec{a} + 3\vec{b} - \vec{c}), B(\vec{a} - 2\vec{b} + 3\vec{c}) & \quad \therefore \overrightarrow{AB} = -\vec{a} - 5\vec{b} + 4\vec{c} \\ C(3\vec{a} + 4\vec{b} - 2\vec{c}) \text{ and } D(k\vec{a} - 6\vec{b} + 6\vec{c}) & \quad \overrightarrow{AC} = -\vec{a} + \vec{b} - \vec{c} \\ & \quad \overrightarrow{AD} = (k-2)\vec{a} - 9\vec{b} + 7\vec{c} \end{aligned}$$

It is given A, B, C, D are coplanar.

$$\begin{vmatrix} -1 & -5 & 4 \\ 1 & 1 & -1 \\ k-2 & -9 & 7 \end{vmatrix} = 0$$

$$(k-2)(5-4) + 9(1-4) + 7(-1+5) = 0$$

$$k-1 = 0$$

$$k = 1$$

35. It is given $n = 8$, $\bar{x} = 25$ and $\sigma = 5$

The mean of x is given as,

$$\bar{x} = \frac{\sum x_i}{n}$$

The value of $\sum x_i$ is given as,

$$\sum x_i = nx = 8 \times 25 = 200$$

$$\frac{1}{n} \sum x_i^2 - \bar{x}^2 = \sigma^2$$

$$\frac{1}{8} \sum x_i^2 - 625 = 25$$

$$\sum x_i^2 = 5200$$

The corrected becomes $\sum x_i^2 = 5200 + 225 + 625 = 6050$

Corrected mean is $\bar{x} = 200 + 15 + 25 = 240$

Corrected variance is $\frac{6050}{10} - \left(\frac{240}{10}\right)^2 = 605 - 576 = 29$

36. Consider the table,

x_i	f_i	Cumulative frequency	$d_i = x_i - 15 $	$f_i d_i$
6	4	4	9	36
9	5	9	6	30
3	3	12	12	36
12	2	14	3	6
15	5	19	0	0
13	4	23	2	8
21	4	27	6	24
22	3	30	7	21
	$N = \sum f_i$ $N = 30$			$\sum f_i d_i = 161$

Here, $\frac{N}{2} = 15$

The cumulative frequency which is just greater than $\frac{N}{2}$ is 19 and corresponding value of x is 15.

So, median = 15.

The mean deviation is,

$$\begin{aligned}\text{Mean deviation} &= \frac{\sum f_i d_i}{N} \\ &= \frac{161}{30} \\ &\approx 5.36\end{aligned}$$

Thus no option is correct.

37. The total possible outcome is,

$$n = 6^3 = 216$$

Here, we choose any 3 numbers out of 6 numbers and place them in ascending order in only one way.

The favorable outcome is,

$$n_f = {}^6C_3 = 20$$

The required probability is,

$$P = \frac{n_f}{n} = \frac{20}{216} = \frac{5}{54}$$

38. It is assumed the events.

$E_1 = 6$ occurs

$E_2 = 6$ does not occur

A = mean reports that it is 6

We have,

$$P\left(\frac{A}{E_1}\right) = \frac{2}{3}$$

$$P\left(\frac{A}{E_2}\right) = \frac{1}{3}$$

The expression for probability that it is actually 6 is,

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1) \times P\left(\frac{A}{E_1}\right)}{P(E_1) \times P\left(\frac{A}{E_1}\right) + P(E_2) \times P\left(\frac{A}{E_2}\right)}$$

$$P\left(\frac{E_1}{A}\right) = \frac{\frac{1}{6} \times \frac{2}{3}}{\left(\frac{1}{6} \times \frac{2}{3}\right) + \left(\frac{5}{6} \times \frac{1}{3}\right)} = \frac{2}{7}$$

The probability that it is not actually 6 is,

$$P = 1 - \frac{2}{7} = \frac{5}{7}$$

The probability that it is actually 5 is,

$$P = \frac{1}{5} \times \frac{5}{7} = \frac{1}{7}$$

39. It is given,

$$P(x = k) = a \left(\frac{k+1}{2^k} \right)$$

We know,

$$\sum P(x = k) = 1$$

$$P(x = 0) + P(x = 1) + P(x = 2) + \\ P(x = 3) + P(x = 4) + P(x = 5) = 1$$

Putting the values in above we get,

$$a + \frac{2a}{2} + \frac{3a}{4} + \frac{4a}{8} + \frac{5a}{16} + \frac{6a}{32} = 1 \\ \frac{16a + 16a + 12a + 8a + 5a + 3a}{16} = 1$$

$$a = \frac{4}{15}$$

The required probability is given as,

$$P(x = 2) + P(x = 3) + P(x = 4) = \frac{3a}{4} + \frac{a}{2} + \frac{3a}{16}$$

$$= \frac{12a + 8a + 3a}{16} = \frac{23a}{16} = \frac{23 \left(\frac{4}{15} \right)}{16} = \frac{23}{60}$$

40. It is given,

$$np = 6$$

$$6 \times q = 2$$

$$q = \frac{1}{3}$$

The value of p becomes,

$$p = 1 - q = 1 - \frac{1}{3} = \frac{2}{3}$$

We have,

$$npq = 2$$

$$n \times \frac{2}{3} \times \frac{1}{3} = 2$$

$$n = 9$$

$$\therefore P(5 \leq x \leq 7) = P(x=5) + P(x=6) + P(x=7)$$

$$P(5 \leq x \leq 7) = {}^9C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^4 + {}^9C_6 \left(\frac{2}{3}\right)^6 \left(\frac{1}{3}\right)^3 + {}^9C_7 \left(\frac{2}{3}\right)^7 \left(\frac{1}{3}\right)^2$$

$$P(5 \leq x \leq 7) = \frac{1}{3^9} \left[\frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} \times 32 + \frac{9 \times 8 \times 7}{3 \times 2 \times 1} \times 64 + \frac{9 \times 8}{2 \times 1} \times 128 \right]$$

$$P(5 \leq x \leq 7) = \frac{1344 + 1792 + 1536}{6561} = \frac{4672}{6561}$$

41. A point P is taken with coordinates of (x, y) , the according to condition,

$$PA^2 + PB^2 = 2PC^2$$

$$(x-2)^2 + (y-3)^2 + (x-3)^2 + (y+6)^2 = 2[(x-5)^2 + (y+7)^2]$$

On solving the above expression we get,

$$10x - 22y = 90$$

By going through options, the points became $(-13, -10)$ which lies on the locus of P .

42. The old coordinates of x is,

$$x = X \cos \theta - Y \sin \theta$$

$$x = 2 \cos 135^\circ - (-6) \sin 135^\circ$$

$$x = \frac{4}{\sqrt{2}} \times \frac{-\sqrt{2}}{-\sqrt{2}} = 2\sqrt{2}$$

The old coordinates of y is,

$$y = X \cos \theta + Y \sin \theta$$

$$y = 2 \cos 135^\circ + (-6) \sin 135^\circ$$

$$y = \frac{8}{\sqrt{2}} \frac{-\sqrt{2}}{-\sqrt{2}} = 4\sqrt{2}$$

The coordinates of point P is $(2\sqrt{2}, 4\sqrt{2})$.

43. The equation of the line is taken as,

$$\frac{x}{a} + \frac{y}{b} = 1$$

The coordinate axes at line meets is $A(a, 0)$ and $B(0, b)$ respectively and its ratio is given as 3:(2)

$$\left(\frac{3 \times 0 + 2 \times a}{3 + 2}, \frac{3 \times b + 2 \times 0}{3 + 2} \right) = \left(\frac{2a}{5}, \frac{3b}{5} \right)$$

Now,

According to give condition at point $(2, -1)$

$$\left(\frac{2a}{5}, \frac{3b}{5} \right) = (2, -1)$$

$$\frac{2a}{5} = 2, \frac{3b}{5} = -1$$

$$a = 5, b = -\frac{5}{3}$$

The required equation of line become,

$$\frac{x}{5} + \frac{y}{\left(-\frac{5}{3}\right)} = 1$$

$$x - 3y = 5$$

$$x - 3y - 5 = 0$$

44. The equation of line passing through the intersection of $2x + y - 4 = 0$ and $2x + y - 4 = 0$ is,

$$(2x + y - 4) + \lambda(x - 3y + 5) = 0$$

$$x(2 + \lambda) + y(1 - 3\lambda) + 5\lambda = 0 \quad \text{(I)}$$

This is at a distance of $\sqrt{5}$ units from the origin.

$$\left| \frac{5\lambda - 4}{\sqrt{(2 + \lambda)^2 + (1 - 3\lambda)^2}} \right| = \sqrt{5}$$

$$\frac{(5\lambda - 4)^2}{4 + \lambda^2 + 4\lambda + 1 + 9\lambda^2 - 6\lambda} = 5$$

$$25\lambda^2 + 30\lambda + 9 = 0$$

On solving the above we get,

$$\lambda = -\frac{3}{5}$$

Substituting the value of λ in equation (I) we get,

$$(2x + y - 4) + -\frac{3}{5}(x - 3y + 5) = 0$$

$$7x + 14y - 35 = 0$$

$$x + 2y - 5 = 0$$

45. The slope of line BC is given as,

$$BC = \left(\frac{0+1}{4-2} \right) = \frac{1}{2}$$

The equation of AD is given as,

$$y - 3 = \frac{-1}{BC}(x + 2)$$

$$y - 3 = \frac{-1}{\left(\frac{1}{2}\right)}(x + 2)$$

$$y - 3 = -2(x + 2)$$

$$2x + y + 1 = 0$$

The slope of line AC is given as,

$$AC = \frac{0-3}{4+2} = \frac{-3}{6} = \frac{-1}{2}$$

The equation of BE is given as,

$$y+1 = \frac{-1}{AC}(x-2)$$

$$y+1 = \frac{-1}{\left(-\frac{1}{2}\right)}(x-2)$$

$$y+1 = 2(x-2)$$

$$y-2x+5=0$$

On solving the above equations we get orthocenter at $(1, -3)$.

$$x, y = \left(\frac{-2+2+4}{3}, \frac{3-1+0}{3} \right)$$

$$x, y = \left(\frac{4}{3}, \frac{2}{3} \right)$$

The equation of line joining $(0, -1)$ and $\left(\frac{4}{3}, \frac{2}{3}\right)$ is,

$$y+3 = \frac{\frac{2}{3}+3}{\frac{4}{3}-1}(x-1)$$

$$y+3 = 11(x-1)$$

$$y+3 = 11x-11$$

$$11x-y-14=0$$

46. The give equation of line is,

$$23x^2 - 48xy + 3y^2 = 0$$

$$3y^2 - 48xy + 23x^2 = 0$$

Here,

$$m_1 + m_2 = 16$$

$$m_1 m_2 = \frac{23}{3}$$

$$\therefore m_1 - m_2 = \sqrt{(m_1 + m_2)^2 - 4m_1 m_2}$$

On putting the value in above expression,

$$m_1 - m_2 = \sqrt{(16)^2 - 4 \times \frac{23}{3}}$$

$$m_1 - m_2 = \frac{26}{\sqrt{3}}$$

On solving,

$$m_1 = 8 + \frac{13}{\sqrt{3}} \text{ and } m_2 = 8 - \frac{13}{\sqrt{3}}$$

The angle of slope is,

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{\left(\frac{26}{\sqrt{3}}\right)}{1 + \frac{23}{3}} = \sqrt{3}$$

$$\theta = 60^\circ$$

The slope of line $2x + 3y + 4 = 0$ is $-\frac{2}{3}$.

The angle between line of slope $8 + \frac{13}{\sqrt{3}}$ and $-\frac{2}{3}$ is,

$$\tan \alpha = \frac{8 + \frac{13}{\sqrt{3}} + \frac{2}{3}}{1 + \left(8 + \frac{13}{\sqrt{3}}\right)\left(\frac{2}{3}\right)} = \sqrt{3}$$

$$\alpha = 60^\circ$$

Thus, the lines form an equilateral triangle.

47. The equation of curve is given as,

$$x^2 + y^2 - xy + 3x + 3y - 2 = 0$$

And,

$$x + 2y = k$$

$$\frac{x + 2y}{k} = 1$$

Substitute the value of 1 in equation of curve as,

$$x^2 + y^2 - xy + 3x\left(\frac{x + 2y}{k}\right) + 3y\left(\frac{x + 2y}{k}\right) - 2\left(\frac{x + 2y}{k}\right)^2 = 0$$

$$x^2(k^2 + 3k - 2) - (k^2 - 9k + 8)xy + (k^2 + 6k - 8)y^2 = 0$$

Since,

$$\angle AOB = 90^\circ$$

$$\therefore k^2 + 3k - 2 + k^2 + 6k - 8 = 0$$

$$2k^2 + 9k - 10 = 0$$

48. The equation of sides of parallelogram is given as,

$$ax^2 + 2hxy + by^2 = 0$$

The equation of diagonal of parallelogram is given as,

$$lx + my = 1$$

The other diagonal is given as,

$$y(bl - hm) = x(am - hl) \dots (I)$$

$$\text{Here, } a = 2, b = 2, h = \frac{3}{2}$$

$$l = -3, m = -1$$

On substitution of value in equation (I) we get,

$$y\left(6 + \frac{3}{2}\right) = x\left(-2 + \frac{9}{2}\right)$$

$$y\left(\frac{15}{2}\right) = x\left(\frac{5}{2}\right)$$

$$3y = x$$

$$x - 3y = 0$$

49. The general equation of circle is given as,

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

The length of tangent drawn to the curve is in the ratio of 2:1

$$\frac{\sqrt{h^2 + k^2 - 2h + 4k - 20}}{\sqrt{h^2 + k^2 - 2h - 8k + 1}} = \frac{2}{1}$$

$$h^2 + k^2 - 2h + 4k - 20 = 4(h^2 + k^2 - 2h - 8k + 1)$$

$$3h^2 + 3k^2 - 6h - 36k + 24 = 0$$

$$h^2 + k^2 - 2h - 12k + 8 = 0$$

Thus, the locus of point P is $x^2 + y^2 - 2x - 12y + 8 = 0$

50. The coordinates of the centre (h, h) of circle touching both coordinate axes is,

$$\therefore \left| \frac{3h - 4h - 12}{\sqrt{(3)^2 + (-4)^2}} \right| = h$$

$$\left| \frac{-h - 12}{5} \right| = h$$

$$-h - 12 = \pm 5h$$

$$h = -2 \text{ or } 3$$

As, $h > 0$.

The equation of circle becomes,

$$(x-3)^2 + (y-3)^2 = 3^2$$

$$x^2 + y^2 - 6x - 6y + 9 = 0$$

51. The pole of the line is taken as (h, k) of the line $9x + y - 28 = 0$ with respect to the circle.

$$x^2 + y^2 - \frac{3}{2}x + \frac{5}{2}y - \frac{7}{2} = 0$$

The equation of polar becomes,

$$hx + ky - \frac{3}{4}(x+h) + \frac{5}{4}(y+k) - \frac{7}{2} = 0$$

$$x\left(h - \frac{3}{4}\right) + y\left(k + \frac{5}{4}\right) - \frac{3}{4}h + \frac{5}{4}k - \frac{7}{2} = 0$$

$$x(4h-3) + y(4k+5) - 3h + 5k - 14 = 0$$

The equation shows the equation of line.

$$\frac{4h-3}{9} = \frac{4k+5}{1} = \frac{-3h+5k-14}{-28} = \lambda$$

$$h = \frac{3+9\lambda}{4}, k = \frac{\lambda-5}{4}, 3h+5k-14 = -28\lambda$$

On simplifying the above we get,

$$-3\left(\frac{3+9\lambda}{4}\right) + 5\left(\frac{\lambda-5}{4}\right) - 14 = -28\lambda$$

$$-22\lambda - 90 = -11\lambda$$

$$90\lambda = 90$$

$$\lambda = 1$$

Thus, the pole of the given line is $(3, -1)$.

52. The centre of the given equation of circle $(x+11)^2 + (y-2)^2 = 225$ and $(x-11)^2 + (y+2)^2 = 25$ is $(-11, 2)$ and $(11, -2)$ and their radius is 15 and 5.

The point of intersection of tangents is given as,

$$P = \left(\frac{11 \times 15 - (-11) \times 5}{15 - 5}, \frac{-2 \times 15 - 2 \times 5}{15 - 5} \right)$$

$$P = \left(\frac{165 + 55}{10}, \frac{-30 - 10}{10} \right) = (22, -4)$$

53. The angle between two circles is given as,

$$\cos \theta = \frac{r_1^2 + r_2^2 - d^2}{2r_1 r_2}$$

Here, the radius is given as r_1 and r_2 and the distance between centre is d .

$$\cos \theta = \frac{(\sqrt{2})^2 + (1)^2 - \left[\sqrt{(2-2)^2 + (1-0)^2} \right]^2}{2 \times \sqrt{2} \times 1}$$

$$(A) \quad \cos \theta = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ \text{ or } 135^\circ$$

$$\cos \theta = \frac{(3)^2 + (\sqrt{17})^2 - \left[\sqrt{(3-2)^2 + (3+2)^2} \right]^2}{2 \times 3 \times \sqrt{17}}$$

$$(B) \quad \cos \theta = 0$$

$$\theta = 90^\circ$$

$$\cos \theta = \frac{(5)^2 + (3)^2 - \left[\sqrt{(-2+2)^2 + (7-0)^2} \right]^2}{2 \times 5 \times 3}$$

$$(C) \quad \cos \theta = \frac{-1}{2}$$

$$\theta = 120^\circ \text{ or } 60^\circ$$

54. The point P is taken as,

$$S_1(a, b) = S_2(a, b)$$

$$a^2 + b^2 + 2ga + 2fb + c = a^2 + b^2 + \frac{3}{2}a + 4b + c = 0$$

$$a\left(2g - \frac{3}{2}\right) + b(2f - 4) = 0$$

This is locus of radical axis.

So, $x\left(2g - \frac{3}{2}\right) + y(2f - 4) = 0$ is the radical axis of the given circles.

This touched the curve of equation $x^2 + y^2 + 2x + 2y + 1 = 0$

The radius of the circle is given as,

$$r = \sqrt{1^2 + 1^2} - 1 = 1$$

The centre of the circle is $(-1, -1)$

The radius of the circle is equal to distance between centre and touching point.

$$\frac{\left|\left(\frac{3}{2} - 2g\right) + (2f - 4)\right|}{\sqrt{\left(\frac{3}{2} - 2g\right)^2 + (2f - 4)^2}} = 1$$

Squaring both sides we get,

$$2\left(\frac{3}{2} - 2g\right)(2f - 4) = 0$$

$$\frac{3}{2} - 2g = 0 \text{ or } 2f - 4 = 0$$

$$g = \frac{3}{4} \text{ or } f = 2$$

55. The equation of line and parabola is,

$$y = 6x + 1$$

$$y^2 = 24x$$

The locus of point of intersection of perpendicular tangent to a parabola is its directrix.

So, the required point is the point of intersection of $y = 6x + 1$ and directrix as $x = -6$.

Thus, its coordinates are $(-6, -35)$.

56. Let $Q(h, k)$ be the point of intersection of the axis AS with the directrix. The point $A(1, 1)$ is the midpoint of QS .

$$\therefore \frac{h+1}{2} = 1 \text{ and } \frac{k-1}{2} = 1$$

$$h = 1 \text{ and } k = 3$$

Thus, the coordinates of the point $Q(1, 3)$.

So, the directrix passes through point $(1, 3)$ and has the gradient of 0.

The equation of directrix is,

$$y - 3 = 0$$

It is assumed the point $p(x, y)$ a point on the parabola and M be the foot of the perpendicular drawn from P on the directrix.

$$PS = PM$$

$$PS^2 = PM^2$$

Substitute the value we get,

$$(x-1)^2 + (y+1)^2 = \left(\frac{y-\sqrt{3}}{\sqrt{1}} \right)^2$$

$$(x-1)^2 + (y+1)^2 = (y-3)^2$$

$$(x-1)^2 = 8(1-y)$$

By going through option a it satisfy the above equation.

Thus, the point $\left(3, \frac{1}{2} \right)$ lies on the parabola $(x-1)^2 = 8(1-y)$

57. The equation of the ellipse is given as,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ and,}$$

$$e = \sqrt{\frac{2}{5}}$$

$$\therefore \sqrt{\frac{a^2 - b^2}{a^2}} = \sqrt{\frac{2}{5}}$$

$$\frac{a^2 - b^2}{a^2} = \frac{2}{5}$$

$$5a^2 - 5b^2 = 2a^2$$

$$a^2 = \frac{5b^2}{3}$$

The ellipse passes through the point $(-3, 1)$.

$$\frac{(-3)^2}{a^2} + \frac{(1)^2}{b^2} = 1$$

$$9b^2 + \frac{5b^2}{3} = \frac{5b^4}{3}$$

$$b^2 = \frac{32}{5}$$

Substituting the value of a and b in general equation we get,

$$\frac{x^2}{\left(\frac{32}{3}\right)} + \frac{y^2}{\left(\frac{35}{5}\right)} = 1$$

$$3x^2 + 5y^2 - 32 = 0$$

58. The equation of tangent is give as,

$$\frac{x}{6} \cos \theta + \frac{y}{3\sqrt{3}} \sin \theta = 1$$

$$3\sqrt{3}x \cos \theta + 6y \sin \theta - 18\sqrt{3} = 0 \dots\dots (I)$$

The product of the length of the perpendiculars from $(3,0)$ and $(-3,0)$, the equation (I) becomes,

$$P = \left| \frac{3 \times 2\sqrt{3} \cos \theta - 18\sqrt{3}}{\sqrt{27 \cos^2 \theta + 36 \sin^2 \theta}} \right| \left| \frac{3 \times 3\sqrt{3} \cos \theta + 18\sqrt{3}}{\sqrt{27 \cos^2 \theta + 36 \sin^2 \theta}} \right|$$

$$P = \frac{36 \times 27 - 9 \times 27 \cos^2 \theta}{36 \sin^2 \theta + 27 \cos^2 \theta} = \frac{9 \times 27 (4 - \cos^2 \theta)}{9 (4 - \cos^2 \theta)} = 27$$

59. The equation of asymptotes is given as,

$$3x + 4y - 2 = 0$$

$$2x + y + 1 = 0$$

The above equation is given by,

$$(3x + 4y - 2)(2x + y + 1) = 0$$

The equation of hyperbola with respect to the above equation is given as,

$$(3x + 4y - 2)(2x + y + 1) = \lambda$$

Here, the constant is given as λ .

The point $(1,1)$ lies on the curve.

$$\therefore (3+4-2)(2+1+1) = \lambda$$

$$5(4) = \lambda = 20$$

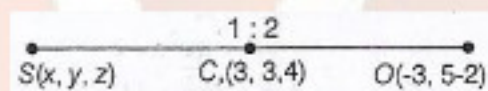
The equation of the hyperbola becomes,

$$(3x+4y-2)(2x+y+1) = 20$$

$$6x^2 + 4y^2 + 11xy - x + 2y - 22 = 0$$

$$6x^2 + 11xy + 4y^2 - x + 2y - 22 = 0$$

60. Consider the figure shown below,



Here, the orthocenter is O , the centroid is G and the circumcentre is S .

$$\left(\frac{-3+2x}{1+2}, \frac{5+2y}{1+2}, \frac{2+2z}{1+2} \right) = (3, 3, 4)$$

$$\left(\frac{-3+2x}{3}, \frac{5+2y}{3}, \frac{2+2z}{3} \right) = (3, 3, 4)$$

$$\frac{-3+2x}{3} = 3, \frac{5+2y}{3} = 3, \frac{2+2z}{3} = 4$$

$$x = 6, y = 2, z = 5$$

$$S(6, 2, 5)$$

61. Assume the equation of the plane is

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

By definition of the centroid $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3} \right) = (6, 6, 3)$

Thus,

$$a = 18$$

$$b = 18$$

$$c = 9$$

The equation of the plane is

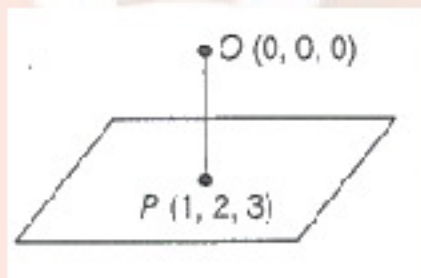
$$\frac{x}{18} + \frac{y}{18} + \frac{z}{9} = 1$$

$$\frac{x + y + 2z}{18} = 1$$

$$x + y + 2z = 18$$

$$x + y + 2z - 18 = 0$$

62. Consider the figure shown below,



DR's of $OP = \langle 1 - 0, 2 - 0, 3 - 0 \rangle = \langle 1, 2, 3 \rangle$

Since OP is perpendicular to the plane, therefore OP is normal to the plane.

The equation of the plane passing through $(1, 2, 3)$ and having $d'r \langle 1, 2, 3 \rangle$ of its normal is

$$1(x-1)+2(y-2)+3(z-3)=0$$

$$x+2y+3z-14=0$$

Thus the $(7, 2, 1)$ lies on the given plane

63. It is given,

$$\lim_{n \rightarrow \infty} \frac{1}{n^3} \{ [1^2 x] + [2^2 x] + [3^2 x] + \dots + [n^2 x] \}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{r=1}^n [r^2 x] = \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{r=1}^n (r^2 x - \{r^2 x\})$$

$$= \lim_{n \rightarrow \infty} \frac{x \times n(n+1)(2n+1)}{n^3 \times 6} - \sum_{r=1}^n \frac{\{r^2 x\}}{n^3}$$

$$= \frac{x}{6} \times 1 \times 2 - 0 = \frac{x}{3}$$

64. It is given,

$$f(x) = \begin{cases} \frac{1 - \sqrt{2} \sin x}{\pi - 4x}, & x \neq \frac{\pi}{4} \\ k & x = \frac{\pi}{4} \end{cases}$$

As $f(x)$ is continuous at

$$x = \frac{\pi}{4}$$

$$f\left(\frac{\pi}{4}\right) = \lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \sqrt{2} \sin x}{\pi - 4x}$$

$$k = \lim_{x \rightarrow \frac{\pi}{4}} \frac{-\sqrt{2} \cos x}{-4}$$

$$k = \frac{\sqrt{2}}{4} \times \frac{1}{\sqrt{2}}$$

$$k = \frac{1}{4}$$

65. It is given,

$$f(x) = x^{\tan^{-1} x}$$

$$\log f(x) = \tan^{-1} x \log x$$

$$\frac{1}{f(x)} \cdot \frac{d}{dx} f(x) = \frac{1}{1+x^2} \log x + \frac{\tan^{-1} x}{x}$$

$$\frac{df(x)}{dx} = x^{\tan^{-1} x} \left[\frac{\log x}{1+x^2} + \frac{\tan^{-1} x}{x} \right]$$

It can also written as

$$g(x) = \sec^{-1} \left(\frac{1}{2x^2 - 1} \right)$$

$$= \cos^{-1} (2x^2 - 1) = 2 \cos^{-1} x$$

$$\frac{d}{dx} g(x) = \frac{-2}{\sqrt{1-x^2}}$$

$$\frac{df}{d \cdot g(x)} = \frac{\frac{df(x)}{dx}}{\frac{dg(x)}{dx}}$$

After solving this equation

$$= \frac{x^{\tan^{-1} x} \left[\frac{\log x}{1+x^2} + \frac{\tan^{-1} x}{x} \right]}{\frac{-2}{\sqrt{1-x^2}}}$$

$$= \frac{-1}{2} \sqrt{1-x^2} \cdot x^{\tan^{-1} x} \left[\frac{\log x}{1+x^2} + \frac{\tan^{-1} x}{x} \right]$$

66. It is given,

$$x = 3\cos t \text{ and } y = 4\sin t$$

$$\frac{x^2}{9} + \frac{y^2}{16} = \cos^2 t + \sin^2 t$$

$$\frac{x^2}{9} + \frac{y^2}{16} = 1$$

Differentiate the above expression

$$\frac{2x}{9} + \frac{2y}{16} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-16}{9} \left[\frac{1 \cdot y - x \frac{dy}{dx}}{y^2} \right]$$

Differentiate the above expression

$$\frac{d^2y}{dx^2} = \frac{-16}{9} \left[\frac{y - x - \left(\frac{-16x}{9y} \right)}{y^2} \right]$$

As $\frac{x^2}{9} + \frac{y^2}{16} = 1$ so the expression is

$$= \frac{-16}{9} \left[\frac{9y^2 + 16x^2}{9y^3} \right] = \frac{-16}{9} \times \frac{144}{9y^3}$$

As the value of $(x_0, y_0) = \left(\frac{3\sqrt{2}}{2}, 2\sqrt{2} \right)$

$$= \frac{-16 \times 144}{81 \times (2\sqrt{2})^3}$$

$$\frac{-16 \times 144}{81 \times 16\sqrt{2}} = \frac{-144}{81 \times \sqrt{2}} = \frac{-16}{9\sqrt{2}} = \frac{-8\sqrt{2}}{9}$$

67. As the given expression is given as

$$y = \frac{2}{\sqrt{a^2 - b^2}} \tan^{-1} \left[\sqrt{\frac{a-b}{a+b}} \tan \frac{x}{2} \right]$$

$$\frac{dy}{dx} = \left[\frac{2}{\sqrt{a^2 - b^2}} \cdot \frac{1}{1 + \left(\frac{a-b}{a+b} \right) \tan^2 \frac{x}{2}} \cdot \sqrt{\frac{a-b}{a+b}} \cdot \sec^2 \frac{x}{2} \cdot \frac{1}{2} \right]$$

$$= \frac{\sec^2 x/2}{a+b} \cdot \frac{a+b}{(a+b) + (a-b) \tan^2 \frac{x}{2}}$$

$$= \frac{\sec^2 x/2}{a+b} \cdot \frac{a+b}{a \tan^2 \frac{x}{2} - b \tan^2 \frac{x}{2}}$$

$$= \frac{\sec^2 x/2}{\left(1 + \tan^2 \frac{x}{2} \right)} \left[a + b \left(\frac{1 - \tan^2 x/2}{1 + \tan^2 x/2} \right) \right]$$

Now the expression will become

$$\frac{dy}{dx} = \frac{1}{a + b \cos x}$$

Differentiate the above expression

$$\frac{d^2y}{dx^2} = \frac{-1}{(a + b \cos x)^2} \cdot (-b \sin x)$$

$$\frac{d^2y}{dx^2} = \frac{b \sin x}{(a + b \cos x)^2}$$

After substitute the values

$$\left. \frac{d^2y}{dx^2} \right|_{x=\frac{\pi}{2}} = \frac{b \sin \pi / 2}{\left(a + b \cos \frac{\pi}{2} \right)^2} = \frac{b}{a^2}$$

68. As the given function is $f'(x) > 0$

$$3x^2 + 2ax + b + 10 \sin x \cos x > 0$$

$$3x^2 + 2ax + b + 5 \sin 2x > 0$$

$$3x^2 + 2ax + b - 5 > 0$$

$$3x^2 + 2ax + (b - 5) > 0$$

As the value of $a > 0$ and $a < 0$

$$(2a)^2 - 4 \times 3 \times (b-5) < 0$$

$$4a^2 - 12(b-5) < 0$$

$$a^2 - 3(b-5) < 0$$

$$a^2 - 3b + 15 < 0$$

Thus no option is correct.

69. Let $y = \cos(x)$, $x = 30^\circ$ and $x + \Delta x = 31^\circ$

The value of

$$(y)_{x=\frac{\pi}{6}} = \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

Then, $\Delta x = 1^\circ = 0.0174$ radian

Consider the given function

$$y = f(x) = \cos(x)$$

Differentiate the above expression with respect to x

$$\frac{dy}{dx} = -\sin(x)$$

$$\left(\frac{dy}{dx}\right)_{x=\frac{\pi}{6}} = -\sin\left(\frac{\pi}{6}\right)$$

$$\left(\frac{dy}{dx}\right)_{x=\frac{\pi}{6}} = -\frac{1}{2}$$

Let Δy be the change in y due to the change Δx in x

$$\begin{aligned}\Delta y &= \frac{dy}{dx} \times \Delta x \\ &= \left(-\frac{1}{2}\right) \times 0.0174 \\ &= (-0.5) \times 0.0174 \approx -0.0087\end{aligned}$$

$$f(36^\circ) = y + \Delta y$$

After solving this

$$\begin{aligned}&= \frac{\sqrt{3}}{2} - 0.0087 \\ &= \frac{1.732}{2} - 0.0087 \\ &= 0.8660 - 0.0087 \\ &= 0.8573\end{aligned}$$

70. Let, $s = x^2 + y^2$

As the value of $x + y = 32$ the expression will be

$$\begin{aligned}&= x^2 + (32 - x)^2 \\ \frac{ds}{dx} &= 2x + 2(32 - x)(-1) \\ &= 2x - 2(32 - x) \\ &= 4x - 64\end{aligned}$$

For maxima and the minima substitute $\frac{ds}{dx} = 0$ in above expression

$$4x - 64 = 0$$

$$x = 16$$

And the value of y is

$$y = 32 - x$$

$$= 32 - 16$$

$$= 16$$

Again after differentiation

$$\frac{d^2x}{dx} = 4 = +ive$$

As s is minimum when $x = y = 16$

The minimum value of

$$s = 16^2 + 16^2$$

$$s = 256 + 256$$

$$s = 512$$

71. Using the mean value theorem we have

$$f'(c) = \frac{f(2) - f(1)}{2 - 1}$$

$$\frac{-14}{(4c-1)^2} = \frac{1 - \frac{5}{3}}{1}$$

$$\frac{-14}{(4c-1)^2} = \frac{-2}{3}$$

$$(4c-1)^2 = 21$$

Now

$$(4c-1) = \pm\sqrt{21}$$

$$\text{As } c = \frac{1 + \sqrt{21}}{4} \notin [1, 2]$$

$$c = \frac{1 + \sqrt{21}}{4}$$

72. Let,

$$\begin{aligned} I &= \int \frac{\sin 2x dx}{\sin^4 x + \cos^4 x} \\ &= \int \frac{2 \sin x + \cos x}{\sin^4 x + \cos^4 x} dx \end{aligned}$$

Dividing by $\cos^2 x$ in numerator and denominator

$$= \int \frac{2 \tan x \sec^2 x}{1 + (\tan^2 x)^2} dx$$

Again let $\tan^2 x = t$

$$I = \int \frac{dt}{1+t^2}$$

$$= \tan^{-1} t + C$$

$$\tan^{-1} (\tan^2 x) + C$$

By comparing with $\tan^{-1} f(x) + C$, we get

$$f(x) = \tan^2 x$$

$$f\left(\frac{\pi}{3}\right) = \left\{ \tan\left(\frac{\pi}{3}\right) \right\}^2 = (\sqrt{3})^2 = 3$$

73. We have,

$$\begin{aligned} I &= \int \left\{ \frac{\log x - 1}{1 + (\log x)^2} \right\}^2 dx \\ &= \int e^t \frac{t^2 + 1 - 2t}{(t^2 + 1)^2} dt \\ &= \int e^t \left\{ \frac{1}{t^2 + 1} + \frac{-2t}{(t^2 + 1)^2} \right\} dt \\ &= \frac{e^t}{t^2 + 1} + C \\ &= \frac{x}{(\log x)^2 + 1} + C \end{aligned}$$

74. It is assumed,

$$\begin{aligned} I &= \int \frac{dx}{x^3 + 3x^2 + 2x} \\ &= \int \frac{dx}{x(x^2 + 3x + 2)} \\ &= \int \frac{dx}{x(x+1)(x+2)} \end{aligned}$$

Let

$$\begin{aligned} \frac{1}{x(x+1)(x+2)} &= \frac{A}{x} + \frac{B}{(x+1)} + \frac{C}{(x+2)} \\ 1 &= A(x+1)(x+2) + Bx(x+2) + Cx(x+1) \end{aligned}$$

Substitute the value of $x = -1$ the value of B will be -1

$$B = -1$$

Substitute $x = -2$ we get

$$C = \frac{1}{2}$$

$$I = \int \left(\frac{1}{2x} - \frac{1}{x+1} + \frac{1}{2(x+2)} \right) dx$$

After differentiate the above expression

$$\begin{aligned}&= \frac{1}{2} \log x - \log(x+1) + \frac{1}{2} \log(x+2) + C \\&= \frac{1}{2} [\log x - 2 \log(x+1) + \log(x+2)] + C \\&= \frac{1}{2} \log \left| \frac{x(x+2)}{(x+1)^2} \right| + C \\&= \frac{1}{2} \log \left| \frac{x^2 + 2x}{(x+1)^2} \right| + C\end{aligned}$$

75. We have,

$$I_n = \int \sec^n x dx$$

$$I_2 = \int \sec^2 x dx$$

$$= \tan x + C_1$$

$$I_4 = \int \sec^4 x dx$$

And

$$= \int \sec^2 x \cdot \sec^2 x dx$$

$$= \int (\tan^2 x + 1) \sec^2 x dx$$

$$= \frac{\tan^3 x}{3} + \tan x + C_2$$

$$\begin{aligned} I_4 - \frac{2}{3}I_2 &= \frac{\tan^3 x}{3} + \tan x + C_2 - \frac{2}{3}\tan x - \frac{2C_1}{3} \\ &= \frac{1}{3}\tan^3 x + \frac{1}{3}\tan x + C \end{aligned}$$

Where $C_2 - \frac{2}{3}C_1 = C$

$$\begin{aligned} &= \frac{1}{3}\tan x(\tan^2 x + 1) + C \\ &= \frac{1}{3}\tan x \sec^2 x + C \end{aligned}$$

76. We have,

$$\begin{aligned} &\lim_{n \rightarrow \infty} \left[\frac{\sqrt{1} + 2\sqrt{2} + 3\sqrt{3} + \dots + n\sqrt{n}}{n^{5/2}} \right] \\ &= \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{r}{n} \right)^{3/2} \cdot \frac{1}{n} \\ &= \int_0^1 x^{3/2} dx = \left[\frac{x^{5/2}}{5/2} \right]_0^1 = \frac{2}{5} \end{aligned}$$

77. Let

$$I = \int_0^{a/3} \frac{f(x)}{f(x) + f\left(\frac{\alpha - 3x}{3}\right)} dx \quad \dots\dots (I)$$

Here $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

$$I = \int_0^{a/3} \frac{f\left(\frac{\alpha}{3} - x\right)}{f\left(\frac{\alpha}{3} - x\right) + f\left(\frac{\alpha - 3\left(\frac{\alpha}{3} - x\right)}{3}\right)} dx$$
$$I = \int_0^{a/3} \frac{f\left(\frac{\alpha - 3x}{3}\right)}{f\left(\frac{\alpha - 3x}{3}\right) + f(x)} dx \quad \dots\dots (II)$$

Add the expression (I) and (II)

$$2I = \int_0^{a/3} 1 dx$$

$$2I = [x]_0^{a/3}$$

$$2I = \frac{\alpha}{3}$$

$$I = \frac{\alpha}{6}$$

78. It is given,

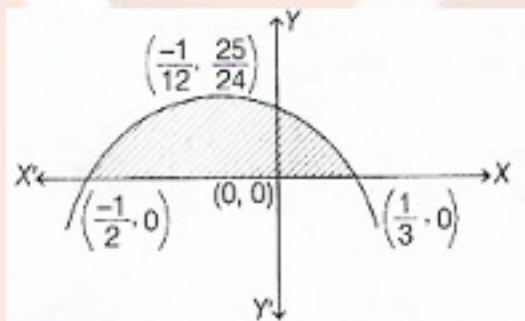
$$y = 1 - x - 6x^2 = -(6x^2 + x - 1)$$

$$y = 6 \left[x^2 + \frac{x}{6} - \frac{1}{6} \right] = -6 \left(x + \frac{1}{12} \right)^2 + \frac{1}{24} + 1$$

$$6 \left(x - \frac{1}{12} \right)^2 = - \left(y - \frac{25}{24} \right)$$

$$\left(x - \frac{1}{12} \right)^2 = - \frac{1}{6} \left(y - \frac{25}{24} \right)$$

The curve is plotted using these points,



The required area is given as,

$$A = \int_{-1/2}^{1/3} y dx = \int_{-1/2}^{1/3} (1 - x - 6x^2) dx$$

$$A = \left[x - \frac{x^2}{2} - 2x^3 \right]_{-1/2}^{1/3} = \left(\frac{18 - 3 - 4}{54} \right) - \left(\frac{-4 - 1 + 2}{8} \right)$$

$$A = \frac{44 + 81}{216} = \frac{125}{216} \text{ sq units}$$

79. The equation of the family of parabolas with focus at the origin and x-axis as its axis is given by

$$\begin{aligned}y^2 &= 4a(x + a) \\&= 4ax + 4a^2\end{aligned}\quad (i)$$

Differentiate the above expression

$$\begin{aligned}2y \frac{dy}{dx} &= 4a \\a &= \frac{1}{2}y \frac{dy}{dx}\end{aligned}\quad (ii)$$

From expression (i) and (ii)

$$y^2 = 2xy \frac{dy}{dx} + y^2 \left(\frac{dy}{dx} \right)^2$$

Here order = $m = 1$ and $n = 2$

Now

$$\begin{aligned}mn - m + n &= 1 \times 2 - 1 + 2 \\&= 3\end{aligned}$$

80. It is given,

$$(1 + e^{x/y}) dx + e^{x/y} \left(1 - \frac{x}{y}\right) dy = 0$$

$$(1 + e^{x/y}) dx = -e^{x/y} \left(1 - \frac{x}{y}\right) dy$$

$$\frac{dx}{dy} = \frac{-e^{x/y} \left(1 - \frac{x}{y}\right)}{(1 + e^{x/y})} \quad \text{.(i)}$$

Substituting $x = vy$ and differentiating it with respect to y .

$$\frac{dx}{dy} = v \frac{dy}{dy} + y \frac{dv}{dy}$$

$$\frac{dx}{dy} = v + y \frac{dv}{dy}$$

Substitute these values in equation (i).

$$v + y \frac{dv}{dy} = \frac{-e^v (1 - v)}{(1 + e^v)}$$

$$y \frac{dv}{dy} = \frac{-e^v (1 - v)}{(1 + e^v)} - v$$

$$y \frac{dv}{dy} = \frac{-(v + e^v)}{(1 + e^v)}$$

$$-\frac{dy}{y} = \left[\frac{1 + e^v}{v + e^v} \right] dv$$

Integrating both sides we get,

$$-\int \frac{dy}{y} = \int \left[\frac{1+e^v}{v+e^v} \right] dv$$

$$\int \frac{1+e^v}{v+e^v} dv = -\log y + \log c$$

Put, $v + e^v = t$
Put, $(1+e^v)dv = dt$

$$\therefore \int \frac{dt}{t} = -\log y + \log c$$

$$\log t = -\log y + \log c$$

$$\log y(v+e^v) = \log c$$

Put v as $\frac{x}{y}$ in above,

$$\log y \left(\frac{x}{y} + e^{x/y} \right) = \log c$$

$$y \left(\frac{x}{y} + e^{x/y} \right) = c$$

$$x + ye^{x/y} = c$$

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Solutions – Physics

81. The equivalent resistance in parallel is,

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{60} + \frac{1}{30}$$

$$R_{eq} = \frac{60 \times 30}{60 + 30}$$

$$R_{eq} = 20 \Omega$$

The tolerance value is calculated as,

$$\Delta R_p = R_{eq} \left[\frac{\Delta R_1}{R_1} + \frac{\Delta R_2}{R_2} - \frac{\Delta R_1 + \Delta R_2}{R_1 + R_2} \right]$$

$$\Delta R_p = 20 \left[\frac{0.36}{60} + \frac{0.09}{30} - \frac{0.36 + 0.09}{30 + 60} \right]$$

$$\Delta R_p = 0.08 \Omega$$

Thus, the resistance in parallel comes out to be 20 ± 0.08 .

82. The expression for the trajectory in projectile motion is given as,

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

The slope of the trajectory is given as,

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{v_y}{v_x}$$

Thus, the assertion is incorrect.

In case of projectile motion, the velocity is always tangent to the path of trajectory. Thus, the reason is correct.

83. The maximum height attained in a projectile motion is given by,

$$h_{\max} = \frac{u^2 \sin^2 \theta}{2g} \quad \dots\dots (1)$$

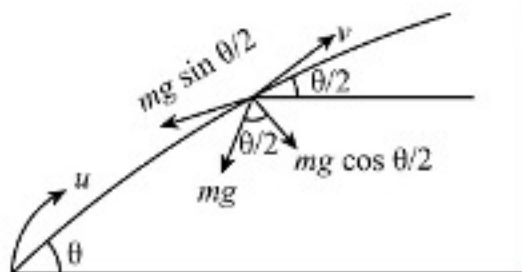
The range in a projectile motion is given by,

$$R = \frac{u^2 \sin 2\theta}{g} \quad \dots\dots (2)$$

The ratio of maximum height to range is,

$$\text{Ratio} = \frac{H_{\max}}{R} = \frac{\frac{u^2 \sin^2 \theta}{2g}}{\frac{u^2 \sin 2\theta}{g}} = \frac{\tan \theta}{4} = \frac{\left(\frac{8}{7}\right)}{4} = \frac{2}{7}$$

84. Consider the figure shown below,



Let the velocity of the projectile at an angle $\frac{\theta}{2}$ be v . The velocity of the projectile in the horizontal direction remains same throughout the motion.

$$v \cos \frac{\theta}{2} = u \cos \theta$$

$$v = \frac{u \cos \theta}{\cos \frac{\theta}{2}}$$

The component of force perpendicular to the direction of motion provides the required centripetal force.

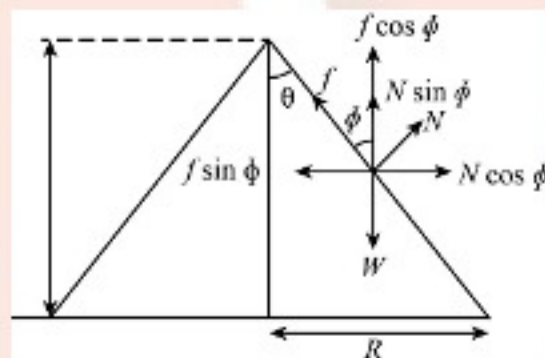
$$\frac{mv^2}{r} = mg \cos \frac{\theta}{2}$$

$$r = \frac{v^2}{g \cos \frac{\theta}{2}}$$

Substituting the value of v in above expression, we get

$$r = \frac{\left(\frac{u \cos \theta}{\cos \frac{\theta}{2}} \right)^2}{g \cos \frac{\theta}{2}} = \frac{u^2 \cos^2 \theta \cdot \sec^3 \left(\frac{\theta}{2} \right)}{g}$$

85. Consider the figure shown below,



Let h be the critical height of sand cone of radius R .

Apply the equilibrium condition in the horizontal direction.

$$f \sin \phi = N \cos \phi$$

$$\mu N \sin \phi = N \cos \phi$$

$$\tan \phi = \frac{1}{\mu} = \frac{R}{h}$$

$$h = \mu R$$

The maximum volume of sand that can be formed over the level of the ground is,

$$V_{\max} = \frac{1}{3} \pi R^2 h$$

$$V_{\max} = \frac{1}{3} \pi R^2 (\mu R) = \frac{\mu \pi R^3}{3}$$

86. The maximum frictional force is calculated as,

$$f = \mu F \cos \theta$$

$$f = 0.25 \times 90 \times \frac{4}{5}$$

$$f = 18 \text{ N}$$

The acceleration of the block is due to the net vertical force in the upward direction.

$$F_v = F \sin \theta - f - mg$$

$$ma = \left(90 \times \frac{3}{5} \right) - 18 - 2 \times 10$$

$$2a = 16 \text{ N}$$

$$a = 8 \text{ m/s}^2$$

87. The work done against the air friction is,

$$W = E_1 - E_2$$

$$W = \frac{1}{2}mv^2 - mgh$$

Substitute the given value in above expression.

$$W = \frac{1}{2} \times 2 \times (8)^2 - 2 \times 10 \times 3$$

$$W = 4 \text{ J}$$

88. The acceleration of the mass pulley system is calculated as,

$$a = \left(\frac{M - m}{M + m} \right) g$$

$$a = \left(\frac{3 \text{ kg} - 2 \text{ kg}}{3 \text{ kg} + 2 \text{ kg}} \right) \times 10 \text{ m/s}^2$$

$$a = \frac{1}{5} \times 10 \text{ m/s}^2$$

$$a = 2 \text{ m/s}^2$$

The distance is given as,

$$s = ut + \frac{1}{2}at^2$$

$$s = 0 + \frac{1}{2} \times 2(2)^2$$

$$s = 4 \text{ m}$$

The required work done is,

$$W = mgs$$

$$W = 3 \times 10 \times 4 = 120 \text{ J}$$

89. Volume increases by 9% means, its radius increases by 1%

Therefore I increases and ω decreases

$$\omega \propto \frac{1}{I} \propto \frac{1}{R^2} \propto \frac{1}{V^{2/3}}$$

The change in angular speed is given as,

$$\frac{\Delta\omega}{\omega} \times 100 = -\frac{2}{3} \times \frac{\Delta V}{V} \times 100$$

$$\frac{\Delta\omega}{\omega} \times 100 = -\frac{2}{3} \times \frac{0.09}{1} \times 100 = -6\%$$

Thus, the decrease in angular speed is 6%.

90. The ratio of tension is given as,

$$Q = \frac{T_1}{T_2} = \frac{mr\omega^2}{(mr\omega^2)_Q + \left(m\frac{r}{2}\omega^2\right)_P}$$

Here, the tension between P and Q is T_1 and the tension in O and P is T_2 .

Substitute the values in above expression.

$$Q = \frac{200 \times 1 \times \omega^2}{(200 \times 1 \omega^2) + \left(200 \times \frac{1}{2} \omega^2\right)} = \frac{200 \times 1 \times \omega^2}{(300 \times 1 \omega^2)} = \frac{2}{3}$$

91. The maximum kinetic energy at an extreme position is given as,

$$(K.E)_{\max} = \frac{1}{2} kx^2$$

$$(T.E - P.E) = \frac{1}{2} kx^2$$

Here, the total energy is $T.E$ and the Potential energy is $P.E$.

Substitute the value in above expression.

$$(9 \text{ J} - 5 \text{ J}) = \frac{1}{2} k (10^{-2})^2$$

$$4 \text{ J} = \frac{1}{2} k 10^{-4}$$

$$k = 8 \times 10^4 \text{ J/m}^2$$

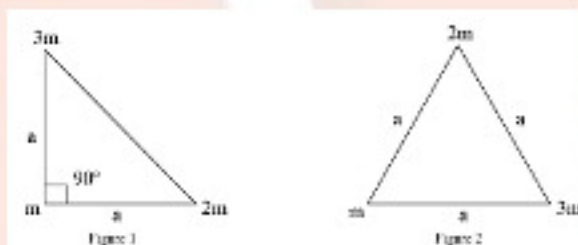
The time period of the oscillator is given as,

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$T = 2\pi\sqrt{\frac{2 \text{ kg}}{8 \times 10^4 \text{ J/m}^2}}$$

$$T = \frac{\pi}{100} \text{ s}$$

92. Consider the figure shown below,



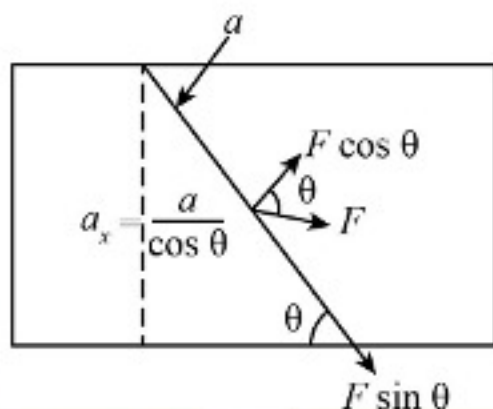
The work done in changing the configuration from one to another is given as,

$$W_{12} = -(U_f - U_i)$$

$$W_{12} = -\left[-\frac{Gm^2}{a}\left(2+3+\frac{6}{\sqrt{2}}\right) - \left(-\frac{Gm^2}{a}(2+3+6)\right)\right]$$

$$W_{12} = -\frac{Gm^2}{a}\left(6 - \frac{6}{\sqrt{2}}\right)$$

93. The figure below shows the free body diagram of the rod,



The vertical distance of the rod is given as,

$$a_x = \frac{a}{\cos \theta}$$

Here, the distance AB is given as a .

The magnitude of shearing stress is given as,

$$\tau = \frac{F \sin \theta}{a_x}$$

Substitute the value in above expression,

$$\tau = \frac{F \sin \theta}{\frac{a}{\cos \theta}}$$

$$\tau = \frac{F}{a} \sin \theta \times \cos \theta$$

94. The density of body in medium is proportional to the acting tension acting on that body.

$$\text{Density of water} - \text{Density of air} \propto T_{\text{water}} - T_{\text{air}}$$

Similarly,

$$\text{Density of liquid} - \text{Density of water} \propto T_{\text{liquid}} - T_{\text{water}}$$

From, the given values,

$$T_{\text{water}} - T_{\text{air}} = T_{\text{liquid}} - T_{\text{water}}$$

Substitute the values in above expression.

$$1000 \text{ kg/m}^3 - 1 \text{ kg/m}^3 = T_{\text{liquid}} - 1000 \text{ kg/m}^3$$

$$T_{\text{liquid}} = 2000 \text{ kg/m}^3$$

95. According to the principle of calorimetry,

$$H_{\text{gain}} = H_{\text{loss}}$$

$$x \times L_v \times (x \times C) \times \Delta T = mC\Delta T + m_w C\Delta T$$

Here, the latent heat of vaporization is L_v , the mass of steam condensed is x and the water equivalent of calorimeter is m_w

Substitute the value in above expression.

$$x \times 540 \times (x \times 1) \times (100 - 90) = 1 \text{ kg} \times 1(90 - 9) + 0.1 \text{ kg} \times 1(90 - 9)$$

$$550x = 89.1$$

$$x = 0.162 \text{ kg}$$

$$x = 162 \text{ g}$$

96. According to "Stefan Boltzmann law",

Heat gain by middle surface is equal to the heat loss by middle surface.

$$\sigma A [(3T)^4 - T_0^4] = \sigma A [T_0^4 - (2T)^4]$$

$$T_0^4 = \frac{97}{2} T^4$$

$$T_0 = \left(\frac{97}{2} \right)^{1/4} T$$

97. It is assumed the n moles of N_2 gas in the cylinder.

The internal energy of the molecules is given as,

$$\frac{1}{2} Mv^2 = C_v \Delta T = \frac{R}{\gamma - 1} \Delta T$$

$$\Delta T = \frac{(\gamma - 1) Mv^2}{2R}$$

According to ideal gas equation,

$$pV = RT$$

$$\Delta p = \frac{R\Delta T}{V}$$

Substitute the values in above expression.

$$\frac{\Delta p}{p} = \frac{\left(\frac{7}{5} - 1\right) \times 28 \times 10^{-3} \times 100^2}{2 \times 8.3 \times 300} = 0.0225 = 2.25\%$$

98. A. The thermal equilibrium of different states is given by Zeroth law of thermodynamics,
B. the first law is also called the energy conservation law.
C. In case of free expansion of gases, the work done is always zero.
D. The second law of thermodynamics states about the direction of flow.

99. The expression for mean free path is given as,

$$\lambda = \frac{1}{\sqrt{2} \pi n d^2}$$

Here, the diameter of gas molecule is d .

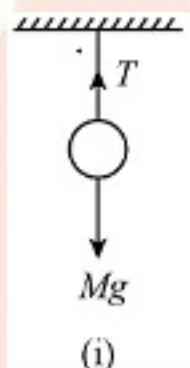
Substitute the given value in above expression.

$$d^2 = \frac{1 \times \pi}{\sqrt{2}\pi \times 2\sqrt{2} \times 10^8 \times 10^{-2}}$$

$$d^2 = \frac{1}{4 \times 10^6}$$

$$d = 5 \times 10^{-4} \text{ cm}$$

100. The figure below shows the state at which car is at rest,



The tension in this case is,

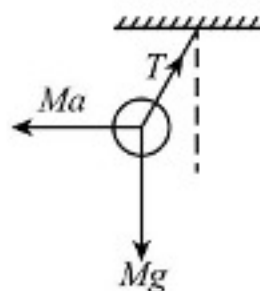
$$T = Mg = mv_1^2$$

$$v_1 = \sqrt{\frac{T}{m}} = \sqrt{\frac{Mg}{m}}$$

Substitute the value in above expression.

$$60 = \sqrt{\frac{T}{m}} = \sqrt{\frac{Mg}{m}}$$

The figure below shows when there is some acceleration in car.



(ii)

The tension in this case is,

$$T_2 = \sqrt{(Ma)^2 + (Mg)^2} = mv_2^2$$

$$v_2 = \frac{[(Ma)^2 + (Mg)^2]^{1/4}}{m^{1/2}}$$

Substitute the value in above expression.

$$66 = \frac{[(Ma)^2 + (Mg)^2]^{1/4}}{m^{1/2}}$$

On dividing both we get,

$$\frac{(a^2 + g^2)^{1/2}}{g} = \frac{121}{100}$$

$$a^2 + g^2 = \left(\frac{121}{100}\right)^2$$

$$a^2 = 146.41 - 100$$

$$a = 6.8 \text{ m/s}^2$$

101. The wavelength of the reflected wave is given as,

$$\lambda_1 = \left(\frac{v - v_s}{v + v_s} \right) \lambda$$

$$\lambda_1 = \left(\frac{v - v_s}{v + v_s} \right) \frac{v}{\nu}$$

Here, the speed of sound in air is v and the frequency of sound wave is 160 Hz.

Substitute the value in above expression.

$$\lambda_1 = \left(\frac{340 - 20}{340 + 20} \right) \frac{340}{160}$$

$$\lambda_1 = \frac{120}{360} \times \frac{340}{160} = \frac{17}{9} \text{ m}$$

102. Consider the figure of equilateral prism.



According to Snell's law for incident ray is,

$$\frac{\sin i}{\sin r_1} = \sqrt{2}$$

Substitute the value in above expression we get,

$$\frac{\sin 0}{\sin r_1} = \sqrt{2}$$

$$\sin r_1 = 0$$

$$r_1 = 0$$

According to Snell's law for emitted ray is,

$$\frac{\sin r_2}{\sin e} = \frac{1}{\mu}$$

$$\sin e = \sqrt{2} \sin r_2$$

The angles of an equilateral triangle is 60° .

$$\sin e = \sqrt{2} \sin 60^\circ$$

$$\sin e = \sqrt{2} \times \frac{\sqrt{3}}{2} > 1$$

Thus, the light is incidenting is more than critical angle and total internally reflected.

The angle of deviation is given as,

$$\delta_d = (i + e) - (r_1 + r_2)$$

Substitute the value in above expression.

$$\delta_d = 0 - 60^\circ = -60^\circ$$

$$|\delta_d| = 60^\circ$$

103. After first polarizer, the unpolarised light become s polarized and its intensity is,

$$I_1 = \frac{I_0}{2}$$

After, 2nd polarizer, the intensity comes out to be,

$$I_2 = \frac{I_0}{2} \cos^2 \theta$$

From Malus law the intensity after 3rd polarizer comes out to be,

$$I_3 = \frac{I_0}{2} \cos^2 \theta \cos^2 (90^\circ - \theta)$$

$$I_3 = \frac{I_0}{2} \cos^2 \theta \sin^2 \theta = \frac{I_0}{8} (2 \cos \theta \times \sin \theta)^2$$

$$I_3 = \frac{I_0}{8} \sin^2 2\theta$$

104. The effective force between the charges is given as,

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 \left[r - t + t\sqrt{K} \right]^2} \dots\dots (i)$$

Here, the dielectric constant of medium is K .

Substitute the values for case I in equation (i)

$$F_1 = \frac{q_1 q_2}{4\pi\epsilon_0 \left[r - \frac{r}{2} + \frac{r}{2}\sqrt{4} \right]^2} = \frac{q_1 q_2}{9\pi\epsilon_0 r^2}$$

Substitute the values for case II in equation (i)

$$F_2 = \frac{q_1 q_2}{4\pi\epsilon_0 \left[r - \frac{r}{3} + \frac{r}{3}\sqrt{3} \right]^2} = \frac{q_1 q_2}{4\pi\epsilon_0 \left(\frac{25}{9} \right) r^2}$$

The ratio of forces is given as,

$$\frac{F_1}{F_2} = \frac{\left(\frac{q_1 q_2}{9\pi\epsilon_0 r^2} \right)}{\left(\frac{q_1 q_2}{4\pi\epsilon_0 \left(\frac{25}{9} \right) r^2} \right)} = \frac{100}{81}$$

105. The force experienced by an electric charge is given as,

$$F = qE$$

$$ma = qE$$

$$a = \frac{qE}{m}$$

Substitute the values in above expression.

$$a = \frac{2 \times 10^{-6} \times 4.2 \times 10^4 \text{ m/s}^2}{0.04}$$

$$a = 2.1 \text{ m/s}^2$$

The expression of time period in absence of an electric field is,

$$T_0 = 2\pi \sqrt{\frac{l}{g}} = 2\pi \sqrt{\frac{l}{10}}$$

The expression of time period in presence of an electric field is,

$$T_1 = 2\pi \sqrt{\frac{l}{a_e}} = 2\pi \sqrt{\frac{l}{a_e}} = 2\pi \sqrt{\frac{l}{a+g}} = 2\pi \sqrt{\frac{l}{12.1}}$$

Here, the effective acceleration on bob is a_e .

The ratio of time period is,

$$\frac{T_0}{T_1} = \sqrt{\frac{12.1}{10}} = \frac{11}{10}$$

$$T_1 = \frac{10}{11} \times \frac{44}{20} = 2 \text{ s}$$

The time taken to complete 15 oscillation is,

$$T_1 = 2 \text{ s} \times 15 = 30 \text{ s}$$

106. The charge stored in presence of air is,

$$q_{air} = C_{air} \times V$$

$$q = 80 \times 30 \times 10^{-6} = 2400 \mu\text{C}$$

The amount of charge present in case of dielectric medium.

$$q_d = C_{dielectric} \times V$$

$$q_d = 1600 \times 30 \times 10^{-6} = 48000 \mu\text{C}$$

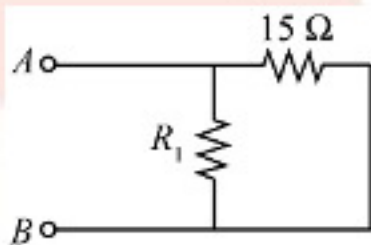
The effective charge, when dielectric is removed is given as,

$$q = q_d - q_{air}$$

$$q = (48 - 2.4) \times 10^{-3} \text{ C}$$

$$q = 45.6 \times 10^{-3} \text{ C}$$

107. Consider the figure shown below,



The expression for total charge in the capacitor is,

$$Q_T = C_1 V_1 + C_2 V_2 + C_3 V_3$$

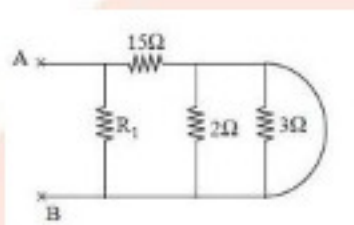
The total capacitance at of the capacitor at point O is given as,

$$C_T = C_1 + C_2 + C_3$$

The potential at O is given as,

$$V = \frac{Q_T}{C_T} = \frac{C_1 V_1 + C_2 V_2 + C_3 V_3}{C_1 + C_2 + C_3}$$

108. Consider the circuit shown below,



Here, the resistance 2Ω and 2Ω are short circuited.

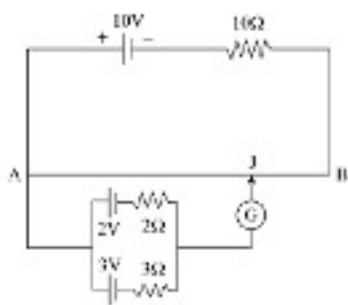
The equivalent resistance is given as,

$$R_{eqt.} = \frac{15R_1}{15 + R_1} = 6$$

$$15R_1 - 6R_1 = 15 \times 6$$

$$R_1 = 10\Omega$$

109. Consider the figure shown below,



The voltage drop in the wire is given as,

$$V' = \frac{V}{(r + R)} \times R$$

$$V' = \frac{10}{20} \times 10 = 5$$

The potential gradient on wire is,

$$x = \frac{V'}{L} = \frac{5}{100} \text{ V/cm}$$

The effective emf in secondary circuit is given as,

$$\frac{1}{r_{\text{eff}}} = \frac{1}{r_1} + \frac{1}{r_2}$$

$$\frac{1}{r_{\text{eff}}} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

$$V = \frac{6}{5}(1+1) = \frac{12}{5}$$

The effective length from point A is given as,

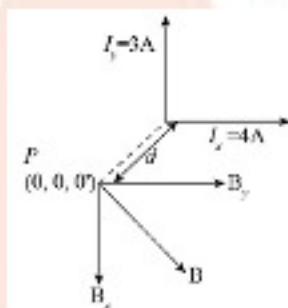
$$l = \frac{V}{x} = \frac{12}{5} \times \frac{100}{5}$$

$$l = 48 \text{ cm}$$

The length from point B comes out to be,

$$l_0 = (100 - 48) \text{ cm} = 52 \text{ cm}$$

110. Consider the figure shown below,



The magnetic field at point P due to I_x is,

$$B_x = -\frac{\mu_0 4}{2\pi d} \hat{j}$$

Here, the negative sign is taken as the direction of magnetic field is in $-y$ direction.

The magnetic field at point P due to I_y is,

$$B_y = \frac{\mu_0 3}{2\pi d} \hat{i}$$

The resultant magnetic field is,

$$B = \sqrt{B_x^2 + B_y^2}$$

Substitute the value in above expression.

$$B = \sqrt{\left(-\frac{\mu_0 4}{2\pi d}\right)^2 + \left(\frac{\mu_0 3}{2\pi d}\right)^2}$$

$$B = \frac{\mu_0}{2\pi d} \sqrt{4^2 + 3^2}$$

$$B = \frac{\mu_0 5}{2\pi d}$$

111. The ratio of current sensitivity is given as,

$$\frac{I_{S_1}}{I_{S_2}} = \frac{\frac{N_1 B_1 A_1}{k_1}}{\frac{N_2 B_2 A_2}{k_2}}$$

Substitute the given value in above expression.

$$\frac{I_{S_1}}{I_{S_2}} = \frac{\frac{30 \times 0.25 \times 4.8 \times 10^{-3}}{1}}{\frac{45 \times 0.5 \times 24 \times 10^{-3}}{1}} = \frac{2}{3}$$

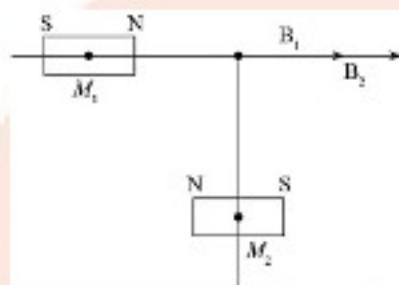
The ratio of voltage sensitivity is given as,

$$\frac{V_{S_1}}{V_{S_2}} = \frac{N_1 B_1 A_1 k_2 R_2}{N_2 B_2 A_2 k_1 R_1}$$

Substitute the value in above expression.

$$\frac{V_{S_1}}{V_{S_2}} = \frac{2}{3} \left(\frac{R_1}{R_2} \right) = \frac{2}{3} \times \frac{14}{10} = \frac{14}{15}$$

112. The figure below shows the direction of magnetic field.



The magnetic field at an axial point is,

$$B_1 = \frac{\mu_0}{4\pi} \times \frac{2M}{r^3} = 10^{-7} \times \frac{2 \times 9}{27 \times 10^{-6}}$$

$$B_1 = \frac{2}{3} \times 10^{-1} \text{ T}$$

The magnetic field at an equatorial point is,

$$B_2 = \frac{\mu_0}{4\pi} \times \frac{M}{r^3} = 10^{-7} \times \frac{9}{27 \times 10^{-6}}$$

$$B_2 = \frac{1}{3} \times 10^{-1} \text{ T}$$

The net effective magnetic field is,

$$B = B_1 + B_2$$

Substitute the value in above expression.

$$B = \left(\frac{2}{3} \times 10^{-1} + \frac{1}{3} \times 10^{-1} \right) \text{T}$$

$$B = 0.1 \text{ T}$$

113. The charge on the plate of the capacitor is,

$$q = CV = C(vBl)$$

Substitute the given value in above expression.

$$q = 10 \times 10^{-6} (2 \times 4 \times 1)$$

$$q = 80 \mu\text{C}$$

114. The expression for impedance in RC circuit is,

$$Z = \sqrt{R^2 + X_C^2} \dots (i)$$

The impedance is given as,

$$Z = 1414 = 1000 \times 1.414 = 1000 \times \sqrt{2}$$

$$Z = 1000\sqrt{1+1} = \sqrt{(1000)^2 + (1000)^2} \dots (ii)$$

On comparing the values we get,

$$X_C = 1000 \text{ R} = 1000 \Omega$$

We know, the capacitive reactance is given as,

$$X_c = \frac{1}{\omega C}$$

$$C = \frac{1}{\omega X_c}$$

Substitute the values in above expression.

$$C = \frac{1}{100 \times 1000} = 10 \mu\text{F}$$

115. The expression for energy density due to electric field is given as,

$$U = \frac{1}{2} \epsilon_0 E^2 = \frac{1}{2} \epsilon_0 \left(\frac{E_0}{\sqrt{2}} \right)^2$$

Substitute the values in above expression.

$$U = \frac{1}{2} 8.85 \times 10^{-12} \left(\frac{40}{\sqrt{2}} \right)^2$$

$$U = 3.54 \times 10^{-9} \text{ J/m}^3$$

116. The energy of photons from state 3 to state 2 is given by,

$$\Delta E = 13.6 \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{13.6 \times 5}{36} = 1.89 \text{ eV}$$

The energy of photons from state 4 to state 2 is given by,

$$\Delta E = 13.6 \left(\frac{1}{4} - \frac{1}{16} \right) = \frac{13.6 \times 12}{64} = 2.55 \text{ eV}$$

The energy of photons from infinite to state 2 is given by,

$$\Delta E = 13.6 \left(\frac{1}{4} - \frac{1}{\infty} \right) = \frac{13.6}{4} = 3.4 \text{ eV}$$

The ratio of kinetic energy of emitted photons is,

$$r = \frac{(2.55 - 1.89) \text{ eV}}{(3.4 - 1.89) \text{ eV}} \approx \frac{7}{16}$$

117. The angular velocity of n^{th} electron is given as,

$$L = mvr = \frac{nh}{2\pi}$$

The de-Broglie wavelength is given as,

$$\lambda = \frac{2\pi r}{n}$$

Here, $r \propto n^2$

$$\therefore \lambda \propto n$$

118. The number of nuclei is given as,

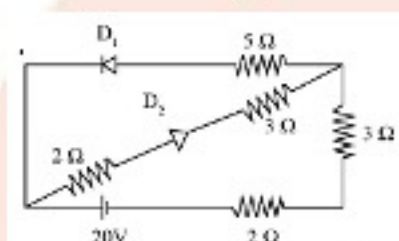
$$N = \frac{E_T}{E_i}$$

Substitute the given value in above expression.

$$N = \frac{1000 \text{ J}}{2000 \text{ MeV}} = \frac{1000 \text{ J}}{2000 \times 1.6 \times 10^{-19} \text{ J}}$$

$$N = 3.125 \times 10^{13}$$

119. Consider the figure shown below,



The diode D_1 is reversed biased and diode D_2 is forward biased.

The voltage drop across is zero as the diode is ideal.

The current through the cell is given as,

$$I = \frac{V}{R_{eff}}$$

Here, the effective resistance is given as R_{eff} .

Substitute the value n above expression.

$$I = \frac{20 \text{ V}}{3 \Omega + 2 \Omega + 3 \Omega + 2 \Omega} = 2 \text{ A}$$

120. The expression for modulation index is,

$$M = \frac{V_{peak, signal}}{V_{peak, carrier}}$$

Substitute the value in above expression.

$$0.6 = \frac{12}{V_{peak, carrier}}$$

$$V_{peak, carrier} = 20 \text{ V}$$

The modulation index in 2nd case is given as,

$$0.75 = \frac{12}{(V_{peak, carrier})_1}$$

$$(V_{peak, carrier})_1 = 16 \text{ V}$$

The change in peak voltage is given as,

$$\Delta V = 20 \text{ V} - 16 \text{ V} = 4 \text{ V}$$

The percentage change is given as,

$$= \frac{4}{20} \times 100\% = 20\%$$

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121. The energy of excited state can be calculated as:

$$r_n = \frac{n^2 \times 527}{Z}$$

$$476.1 = \frac{n^2 \times 527}{1}$$

$$n^2 = 9$$

$$n = 3$$

We know that,

$$E_n = -2.18Z^2 \times 10^{-18} \text{ J} \times \frac{Z^2}{n^2}$$

For hydrogen Z is equal to 1. Therefore,

$$E_3 = -\frac{2.18 \times 10^{-18} \text{ J}}{9}$$

$$= -2.42 \times 10^{-19} \text{ J}$$

122. First case:

$$h(1.6 \times 10^6 - \nu_0) = 2K.E \dots\dots (I)$$

Second case:

$$h(1 \times 10^6 - \nu_0) = K.E \dots\dots (II)$$

By dividing both the equations, we get

$$\frac{h(1.6 \times 10^6 - \nu_0)}{h(1.0 \times 10^6 - \nu_0)} = \frac{2K.E}{K.E}$$

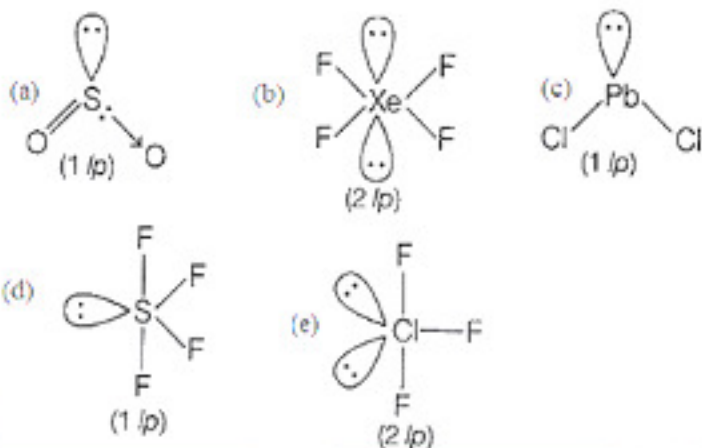
$$1.6 \times 10^6 - \nu_0 = 2 \times 1.0 \times 10^6 - \nu_0$$

Simplify the above equation;

$$\nu_0 = 4 \times 10^{15} \text{ Hz}$$

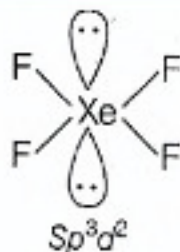
123. The given electronic configuration of an element in -2 oxidation state is $1s^2 2s^2 2p^6 3s^2 3p^6$. There are total 16 electrons which means the compound is sulphur which belongs to group 16 and period 3. The oxidation state of -2 is shown by sulphur.

124. The count of lone pair of the given species is shown below.

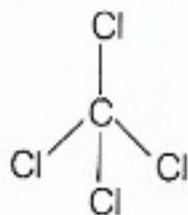


Therefore, SO_2 , PbCl_2 and SF_4 has one lone pair.

125. XeF_4 has a hybridization of sp^3d^2 because it contains 4σ electrons and 2 lone pair of electrons. Its shape is square planar. The structure is shown below.



CCl_4 has a hybridization of sp^3 because it contains 4σ bonds. Its shape is tetrahedral geometry. The structure is shown below.



Tetrahedral geometry

126. It is given that mixture of 16 g of H_2 , He and O_2 exert a pressure of 10 atm.

Therefore,

$$\begin{aligned}\text{Mixture of 36 g exert pressure} &= \frac{36}{10} \\ &= 3.6\end{aligned}$$

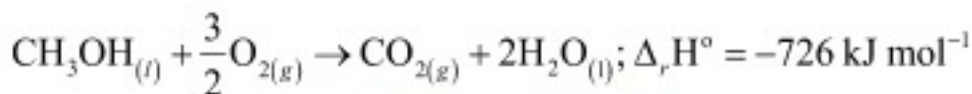
127. The total molarity is equal to

$$\begin{aligned}M &= \frac{M_1V_1 + M_2V_2}{V_3} \\ &= \frac{0.15 \times 1 + 0.2 \times 0.5}{1.5} \\ &= 0.15\end{aligned}$$

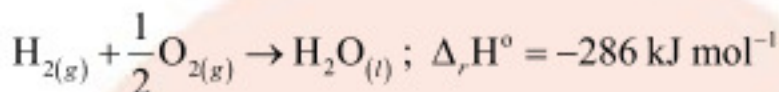
The remaining moles of $\text{K}_2\text{Cr}_2\text{O}_7$ is equal to

$$0.2 - 0.15 = 0.05 \text{ mol/L}$$

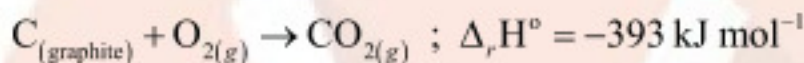
128. The reaction (I) is shown below.



The reaction (II) is shown below.



The reaction (III) is shown below.

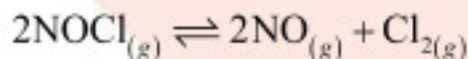


Multiply the reaction (III) by 2.

From reaction (II) + (III) – (I), we can obtain the standard enthalpy formation of $\text{CH}_3\text{OH}_{(l)}$

$$\begin{aligned} -286 \times 2 - 393 - (-726) &= \Delta H_f^\circ \\ \Delta H_f^\circ &= -239 \text{ kJ/mol} \end{aligned}$$

129. The given reaction is:



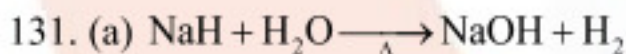
$$\begin{aligned} \Delta n &= \text{Moles of gaseous products} - \text{Moles of gaseous reactants} \\ &= 3 - 2 \\ &= 1 \end{aligned}$$

Now, the value of K_c is calculated as:

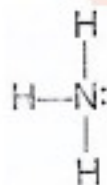
$$\begin{aligned}
 K_p &= K_c (RT)^{\Delta n} \\
 &= 4 \times 10^{-6} \text{ mol/L} \left(0.0831 \text{ L bar K}^{-1} \text{ mol}^{-1} \times 1000 \text{ K} \right)^2 \\
 &= 3.32 \times 10^{-4} \text{ bar}
 \end{aligned}$$

130. The pH is calculated as:

$$\begin{aligned}
 \text{pH} &= 7 + \frac{1}{2} [\text{p}K_a - \text{p}K_b] \\
 &= 7 + \frac{1}{2} [4.76 - 3.26] \\
 &= 7 + \frac{1}{2} [1.50] \\
 &= 7.75
 \end{aligned}$$

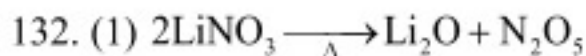


(b) NH_3 is an example of electron rich hydride due to the availability of lone pairs as shown below.

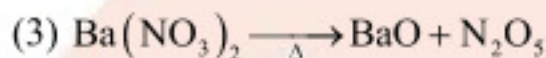
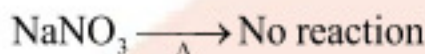


(c) Saline hydride is not formed by nickel.

Therefore, statements (a) and (b) are correct.

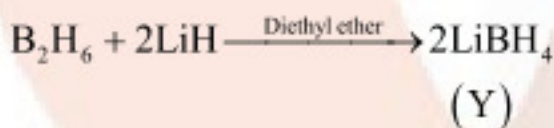
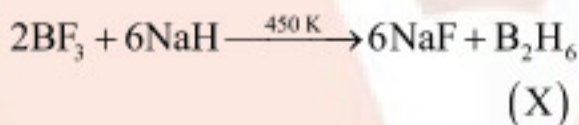


(2) Sodium nitrate does not give oxide on heating as it is thermally stable as shown below.



Therefore, NaNO_3 is a nitrate which does not give its oxide.

133. The reactions are shown below.

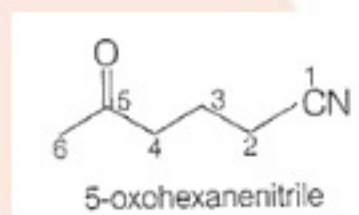


134. The formation of $[\text{SiF}_6]^{2-}$ is possible but $[\text{SiCl}_6]^{2-}$ is not possible due to greater size of chlorine than that of fluorine. The anions which are having large size cannot produce complex with silicon.

Therefore, both Assertion (A) and Reason (R) are correct but (R) is not the correct explanation of (A)

135. Nowadays, the environment friendly chemical is hydrogen peroxide which is used for bleaching the paper in the presence of a suitable catalyst.

136. The IUPAC name of the given structure is 5-oxohexanenitrile as shown below.

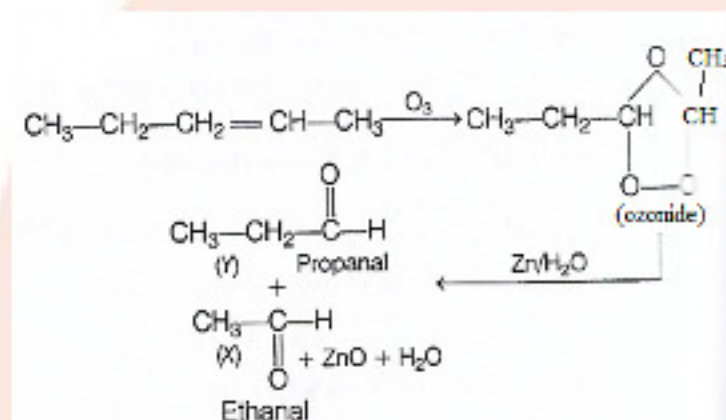


137. a. The less pollution is caused by petrol and CNG operated automobiles.
- b. Alkanes which are having tertiary hydrogen (3°) can be oxidized to alcohols by KMnO_4
- c. Methane cannot be prepared by Kolbe's electrolytic method as methane is having only one carbon atom. This method is used to produce ethane and higher members.

d. When alkyl chloride undergoes reduction with zinc and HCl, it produces alkane.

Therefore, statements (a), (b) and (d) are correct.

138. The products (X) and (Y) formed are shown below.



139. The total number of body centered lattices is 3 which are simple cubic, tetragonal and orthorhombic.

140. The osmotic pressure is calculated as:

$$\pi = iCRT$$

$$\pi = i \frac{m}{M \times V} RT$$

Substitute the values

$$3.735 = i \times \frac{17.4 \times 0.083 \times 300}{174 \times 2}$$

$$i = 3$$

141. The total mass is $1000 + 120 = 1120$ g

Volume (V) is calculated as:

$$d = \frac{m}{V}$$

$$1.12 = \frac{1120}{V}$$

$$V = 1000 \text{ mL}$$

The molarity is calculated as:

$$\begin{aligned} \text{Molarity (C)} &= \frac{w}{M} \times \frac{1000}{V} \\ &= \frac{120}{60} \times \frac{1000}{1000} \\ &= 2 \end{aligned}$$

142. The reaction at anode is as follows:



The reaction at cathode is shown below.



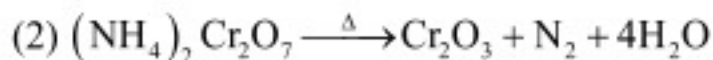
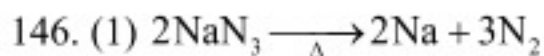
143. For a first order reaction, time can be calculated as:

$$\begin{aligned}K &= \frac{2.303}{t} \log \frac{a}{a-x} \\t &= \frac{2.303}{K} \log \frac{a}{a-x} \\&= \frac{2.303}{4.606 \times 10^{-3}} \log \frac{1}{1-9} \\&= 500 \text{ s}\end{aligned}$$

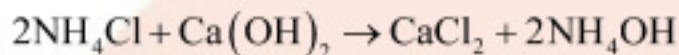
144. A mixture of gases is formed when a mixture of N_2 and O_2 react with each other at room temperature. They do not form aerosol. Therefore, statement (a) is not correct.

145. In Ellingham diagram, the plot is drawn between temperature and ΔG° as shown below.



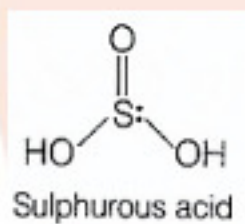


(3) The reaction of ammonium chloride with calcium hydroxide form CaCl_2 instead of N_2 as shown below.



Therefore, reaction (3) does not liberate N_2

147. Sulphurous acid that is H_2SO_3 contains lone pair of electrons as follows:



148. Among all noble gases, helium has lowest boiling point as its atomic size is very small and surface area is also small.

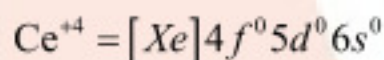
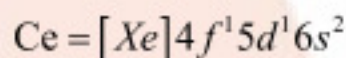
Therefore, statement (4) is not correct.

149. The relation between octahedral splitting energy (denoted by Δ_0) and tetrahedral splitting energy (denoted by Δ_t) is as follows:

$$\Delta_t = \frac{4}{9} \Delta_0$$

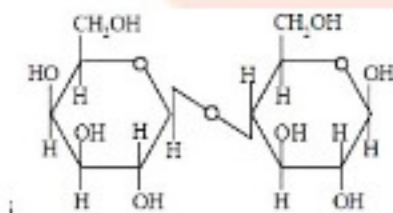
$$9\Delta_t = 4\Delta_0$$

150. In Lanthanide series, Cerium is the element which is well known to exhibit +4 oxidation state.



151. In ionic polymerization, alkyl lithium (R-Li) acts as effective chain initiator

152. The structure of lactose is shown below.

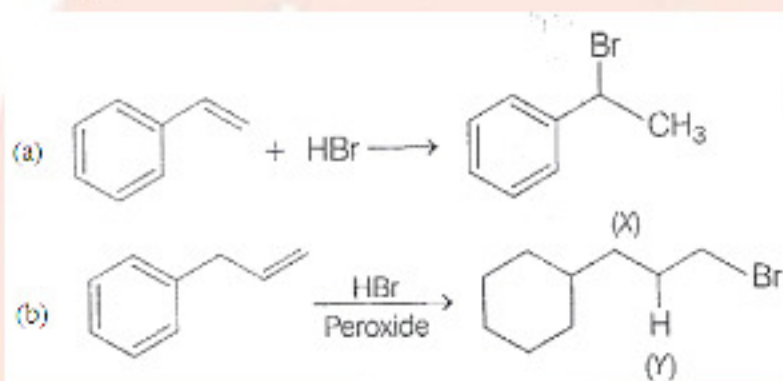


153. a) Agonists are those drugs which imitate natural messenger by switching on the receptor.

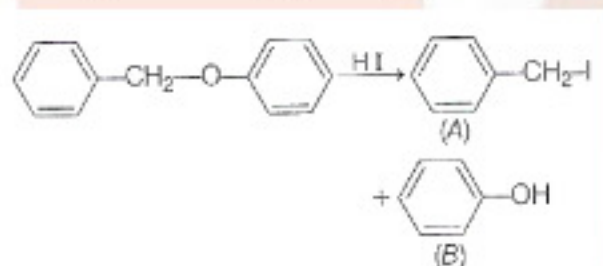
d) An example of antihistamine is seldane.

Therefore statements (a) and (d) are correct.

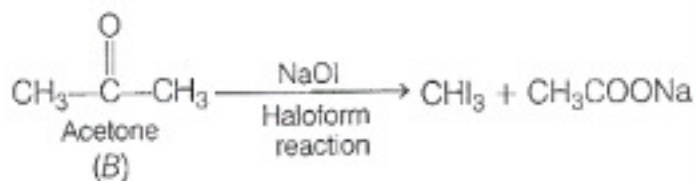
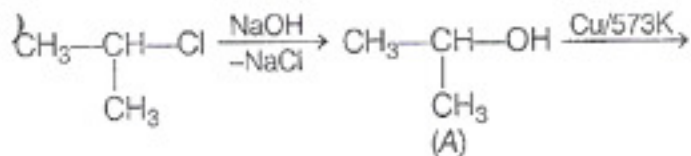
154. The products (X) and (Y) formed are shown below.



155. The products (A) and (B) formed are shown below.

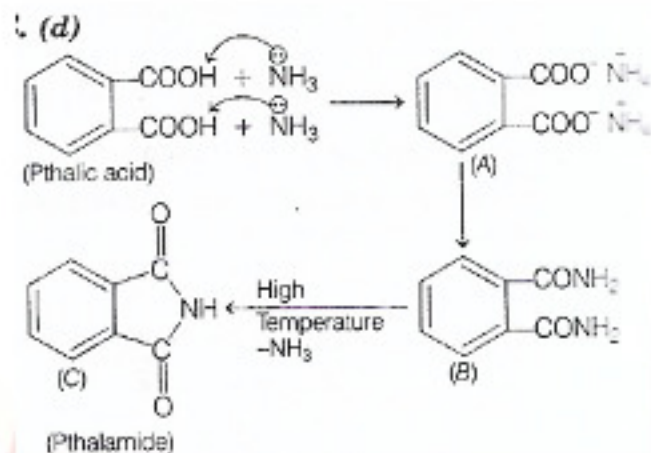


156. The products (A), (B) and (C) formed are shown below

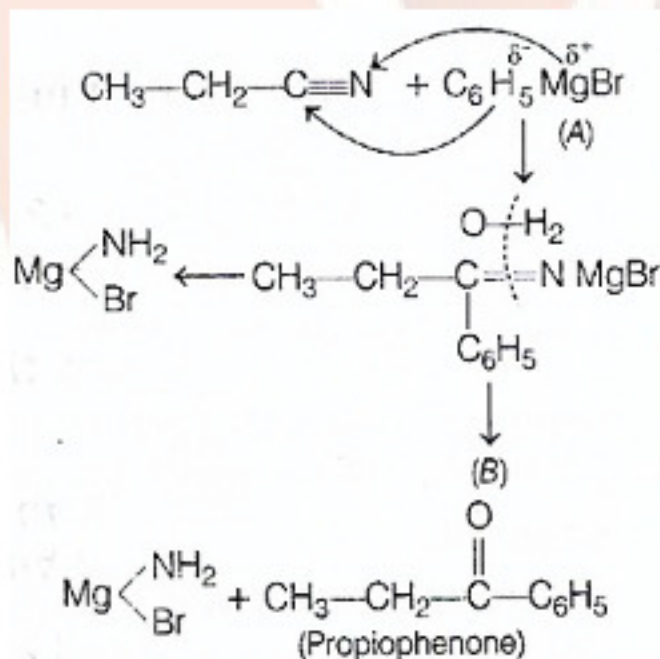


157. (A) Lucas reagent is composed of anhydrous ZnCl_2 and conc. HCl . This reagent is used to differentiate between primary, secondary and tertiary alcohols.
- (B) Clemmensen reagent is composed of anhydrous $\text{Zn} - \text{Hg}/\text{HCl}$.
- (C) Tollens' reagent is composed of $[\text{Ag}(\text{NH}_3)_2]^+$
- (D) In Stephen reaction, reagents such as SnCl_2 and HCl are used to reduce compounds of nitrogen.
- Therefore, A-III, B-IV, C-II and D-I

158. The products (A), (B) and (C) formed are shown below



159. The products (A) and (B) formed are follows:



160. The products (X), (Y) and (Z) formed are follows:

